

# Exploratory Data Analysis on

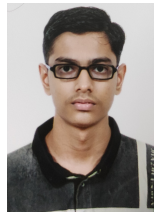
SHORT DURATION DEBT FUNDS

by

## Group 3



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Course Code: IT 462  
Semester: Autumn 2025

---

Under the guidance of

**Dr. Gopinath Panda**



Dhirubhai Ambani University  
November 30, 2025

# ACKNOWLEDGMENT

I am writing this letter to express my heartfelt gratitude for your guidance and support throughout the duration of my project titled SHORT DURATION DEBT FUNDS. Your invaluable assistance has played a pivotal role in shaping the successful completion of this endeavor.

I am extremely fortunate to have had the opportunity to work under your mentorship. Your expertise, encouragement, and willingness to share your knowledge have been instrumental in elevating the quality and scope of my project. Your constructive feedback and insightful suggestions have helped me overcome challenges and develop a deeper understanding of the subject matter.

Furthermore, I would like to extend my appreciation to the entire team at Dhirubhai Ambani University for fostering an environment of collaboration and innovation. The resources and facilities provided have been crucial in conducting comprehensive research and analysis.

I would also like to express my gratitude to my peers and colleagues who have been supportive throughout this journey. Their valuable input and camaraderie have been a constant source of motivation.

Completing this project has been a tremendous learning experience, and I am confident that the knowledge and skills acquired during this endeavor will serve as a solid foundation for my future endeavors.

Once again, thank you for your unwavering guidance and belief in my abilities. Your mentorship has been invaluable, and I am truly grateful for the opportunity to work with you.

Sincerely,

Chirag Katkoriya: 202301259, Pranav Bavadiya: 202301458, Om Patel: 202301027

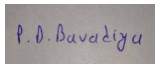
# DECLARATION

We, as Group-3 hereby declare that the EDA project work presented in this report is our original work and has not been submitted for any other academic degree. All the sources cited in this report have been appropriately referenced.

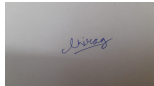
We acknowledge that the data used in this project is obtained from the AMFI site. We also declare that we have adhered to the terms and conditions mentioned in the website for using the dataset. We confirm that the dataset used in this project is true and accurate to the best of our knowledge.

We acknowledge that we have received no external help or assistance in conducting this project, except for the guidance provided by our mentor Prof. Gopinath Panda. We declare that there is no conflict of interest in conducting this EDA project.

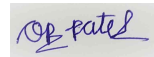
We hereby sign the declaration statement and confirm the submission of this report on 30th November, 2025.



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# CERTIFICATE

This is to certify that Group 3 comprising Chirag Katkoriya, Pranav Bavadiya, and Om Patel has successfully completed an exploratory data analysis (EDA) project on Short-term debt funds, which was obtained majority from [www.amfiindia.com](http://www.amfiindia.com)

The EDA project presented by Group 3 is their original work and has been completed under the guidance of the course instructor, Prof. Gopinath Panda, who has provided support and guidance throughout the project. The project is based on a thorough analysis of the Short Duration Debt Funds dataset, and the results presented in the report are based on the data obtained from the dataset.

This certificate is issued to recognize the successful completion of the EDA project on the Short Duration Debt Funds, which demonstrates the analytical skills and knowledge of the students of Group 3 in the field of data analysis.

Signed,  
Dr. Gopinath Panda,  
IT 462 Course Instructor  
Dhirubhai Ambani Institute of Information and Communication Technology  
Gandhinagar, Gujarat, INDIA.

November 30, 2025

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## **Abstract**

This project is an Exploratory Data Analysis (EDA) of Short Duration Debt Mutual Funds in India, using data sourced from AMFI (Association Of Mutual Funds in India) and publicly available fund details of banks like SBI (State Bank of India), Axis and HDFC Bank. This is very important for investors who try to get a balance between liquidity and yield. The dataset has some essential financial performance and attributes of various schemes, including Scheme Name, Assets Under Management (AUM), Yield to Maturity (YTM). The objective is to understand trends in fund performance with the help of Riskometer, Rating, Net Asset Value and many other columns. The analysis involves data cleaning, handling missing values, imputation, outlier detection, followed by comprehensive univariate and multivariate visualizations.

# Chapter 1. Introduction

## 1.1 Your Project idea

Our Topic is Short Duration Debt Fund. The idea of this project is to perform a detailed Exploratory Data Analysis on Short-term debt mutual funds in India, using data from AMFI (Association of Mutual Funds in India). These are fixed income schemes that invest for a duration of generally 1-3 years, making them suitable for investors seeking good returns with controlled risk.

We want to understand the performance, risk patterns and characteristics of Short Duration Debt Funds through statistical summaries, visualizations, analysis and relation among different parameters. The dataset considered for this project consists of some attributes such as Scheme Name, Net Asset Value (NAV), Riskometer, CRISIL Rating and Launch Date. These variables allow us to study how NAVs fluctuate over time, how ratings correlate with fund performance, whether risk categories align with actual behaviors.

We have done missing value analysis, dataset visualizations after outlier analysis. We have also done feature engineering to make a predictive trained model. We have predicted Risks based on the information of the debt funds.

## 1.2 Data Collection

We searched across major online sites related to Debt Funds such as Money Control, AMFI India, Value Research Online.

We used [AMFI India Website](https://www.amfiindia.com/otherdata/transaction-in-debt) and from this we got the Net Asset Value History.

<https://www.amfiindia.com/otherdata/transaction-in-debt>

We got our major dataset Schemes from the following site and also of Axis Bank

<https://www.amfiindia.com/net-asset-value>

We got SBI Data from the following link

<https://dhan.co/mutual-funds/sbi-short-term-debt-fund-direct-growth>

Then after this, in majority of sites, there was an option to download the excel file directly from the site, and hence we were able to collect it and then we have pushed the dataset into the Github account.

[https://raw.githubusercontent.com/Pranav-Bavadiya/EDA\\_Dataset\\_03/main/Dataset/EDA\\_Project\\_Dataset-4.xlsx](https://raw.githubusercontent.com/Pranav-Bavadiya/EDA_Dataset_03/main/Dataset/EDA_Project_Dataset-4.xlsx)

We go the main\_funds dataset from the following:

<https://www.moneycontrol.com/mutual-funds/performance-tracker/returns/short-duration-funds.html>

## 1.3 Dataset Description

Main\_Funds These are the 26 main mutual funds which are short term debt funds, with 20 columns.

|    | Scheme Name                                        | Plan        | Category Name       | Crisil Rating | AuM (Cr) | YTM  | 1W     | 1M     | 3M     | 6M     | YTD    | 1Y     | 2Y     | 3Y     | 5Y     | 10Y    | Riskometer | Expense Ratio (%) | Net Assets (Cr) | Launch Date |
|----|----------------------------------------------------|-------------|---------------------|---------------|----------|------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|------------|-------------------|-----------------|-------------|
| 0  | Aditya Birla Sun Life Short Term Fund - Direct...  | Direct Plan | Short Duration Fund | 1.0           | 10696.23 | 7.21 | 0.0018 | 0.0080 | 0.0149 | 0.0358 | 0.0758 | 0.0884 | 0.0884 | 0.0835 | 0.0691 | 0.0807 | Moderate   | 0.37              | 10696.0         | 2013-01-01  |
| 22 | Tata Short Term Bond Fund - Direct Plan - Growth   | Direct Plan | Short Duration Fund | 3.0           | 3305.01  | 6.85 | 0.0014 | 0.0059 | 0.0120 | 0.0299 | 0.0702 | 0.0817 | 0.0833 | 0.0805 | 0.0624 | 0.0686 | Moderate   | 0.34              | 3305.0          | 2013-01-01  |
| 16 | LIC MF Short Duration Fund - Direct Plan - Growth  | Direct Plan | Short Duration Fund | 3.0           | 151.13   | 6.76 | 0.0015 | 0.0057 | 0.0123 | 0.0340 | 0.0743 | 0.0860 | 0.0866 | 0.0813 | 0.0612 | NaN    | Moderate   | 0.36              | 151.0           | 2019-02-01  |
| 4  | Baroda BNP Paribas Short Duration Fund - Direct... | Direct Plan | Short Duration Fund | 2.0           | 271.94   | 6.96 | 0.0012 | 0.0063 | 0.0133 | 0.0346 | 0.0754 | 0.0870 | 0.0864 | 0.0830 | 0.0652 | 0.0771 | Moderate   | 0.44              | 272.0           | 2013-01-01  |
| 24 | Union Short Duration Fund - Direct Plan - Growth   | Direct Plan | Short Duration Fund | NaN           | 363.64   | 6.87 | 0.0014 | 0.0069 | 0.0137 | 0.0355 | NaN    | NaN    | NaN    | NaN    | NaN    | NaN    | Moderate   | 0.33              | 364.0           | 2025-01-31  |

Figure 1.1: Main Funds Sample

This table give good financial details about dataset's size, performance, risk profile and various return percentages over different periods of time along with AuM value, which is mainly sourced from AMFI site.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 26 entries, 0 to 25
Data columns (total 20 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Scheme Name                          26 non-null    object
1   Plan                                26 non-null    object
2   Category Name                        26 non-null    object
3   Crisil Rating                        24 non-null    float64
4   AuM (Cr)                            25 non-null    float64
5   YTM                                  25 non-null    float64
6   1W                                   25 non-null    float64
7   1M                                   25 non-null    float64
8   3M                                   25 non-null    float64
9   6M                                   25 non-null    float64
10  YTD                                  24 non-null    float64
11  1Y                                   24 non-null    float64
12  2Y                                   24 non-null    float64
13  3Y                                   23 non-null    float64
14  5Y                                   20 non-null    float64
15  10Y                                  18 non-null    float64
16  Riskometer                           26 non-null    object
17  Expense Ratio (%)                     25 non-null    float64
18  Net Assets (Cr)                       25 non-null    float64
19  Launch Date                           25 non-null    datetime64[ns]
dtypes: datetime64[ns](1), float64(15), object(4)
memory usage: 4.2+ KB
```

Figure 1.2: Main Funds Info

We got statistical parameters like count, mean, minimum value, maximum value, 25 percentile, 50 percentile, 75 percentile, standard deviation using describe function.

|       | Crisil Rating | AUM (Cr)     | YTM       | 1M        | 1M        | 3M        | 6M        | YTD       | 1Y        | 2Y        | 3Y        | 5Y        | 10Y       | Expense Ratio (%) | Net Assets (Cr) | Launch Date         |
|-------|---------------|--------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-------------------|-----------------|---------------------|
| count | 24.000000     | 25.000000    | 25.000000 | 25.000000 | 25.000000 | 25.000000 | 25.000000 | 24.000000 | 24.000000 | 24.000000 | 23.000000 | 20.000000 | 18.000000 | 25.000000         | 25.000000       | 25                  |
| mean  | 2.958333      | 5412.504400  | 6.868000  | 0.001380  | 0.006160  | 0.013452  | 0.034108  | 0.073808  | 0.085392  | 0.08575   | 0.081317  | 0.066780  | 0.073600  | 0.348400          | 5412.440000     | 2015-06-05 03:50:24 |
| min   | 1.000000      | 71.330000    | 6.260000  | 0.000600  | 0.004500  | 0.010300  | 0.028600  | 0.066500  | 0.076400  | 0.07950   | 0.076300  | 0.059000  | 0.062300  | 0.200000          | 71.000000       | 2013-01-01 00:00:00 |
| 25%   | 2.000000      | 195.110000   | 6.730000  | 0.001300  | 0.005600  | 0.013000  | 0.033300  | 0.072600  | 0.084025  | 0.08485   | 0.080350  | 0.062050  | 0.070350  | 0.300000          | 195.000000      | 2013-01-01 00:00:00 |
| 50%   | 3.000000      | 925.170000   | 6.870000  | 0.001400  | 0.006300  | 0.013600  | 0.034100  | 0.074250  | 0.086100  | 0.08635   | 0.081500  | 0.064150  | 0.074050  | 0.350000          | 925.000000      | 2013-01-01 00:00:00 |
| 75%   | 4.000000      | 10673.190000 | 7.000000  | 0.001500  | 0.006700  | 0.014300  | 0.035500  | 0.075725  | 0.087175  | 0.08715   | 0.083000  | 0.067500  | 0.077375  | 0.380000          | 10673.000000    | 2018-03-16 00:00:00 |
| max   | 5.000000      | 22387.340000 | 7.310000  | 0.001900  | 0.008000  | 0.015200  | 0.037300  | 0.078700  | 0.091800  | 0.09030   | 0.084900  | 0.106200  | 0.082200  | 0.520000          | 22387.000000    | 2025-01-31 00:00:00 |
| std   | 1.082636      | 7028.063679  | 0.223737  | 0.000274  | 0.000834  | 0.001218  | 0.002076  | 0.002956  | 0.003497  | 0.00272   | 0.002152  | 0.010116  | 0.005355  | 0.072266          | 7028.129683     | NaN                 |

Figure 1.3: Main Funds Describe

Schemes This table gives a good overview of several Short Duration Debt Funds and it showcases Net Asset Value, Riskometer, Rating and Dates.

|     | Scheme NAV Name                                   | Net Asset Value | Date       | Rating | Riskometer | Launch Date         |
|-----|---------------------------------------------------|-----------------|------------|--------|------------|---------------------|
| 186 | NIPPON INDIA SHORT DURATION FUND - IDCW Option    | 22.0329         | 2025-11-07 | 4.0    | Moderate   | NaN                 |
| 90  | Franklin India Short Term Income Plan- Retail ... | 0.0000          | 2020-07-20 | NaN    | Very High  | NaN                 |
| 135 | Invesco India Short Duration Fund - Daily IDCW... | 1036.8661       | 2025-11-07 | 3.0    | Moderate   | 2013-01-01 00:00:00 |
| 14  | Axis Short Duration Fund - Regular Plan - Week... | 10.4081         | 2025-11-07 | 4.0    | Moderate   | NaN                 |
| 84  | Franklin India Short Term Income Plan- Retail ... | 0.0000          | 2025-11-07 | NaN    | Very High  | NaN                 |

Figure 1.4: Schemes Samples

There are in total 249 entries in this dataset Schemes, and 6 categories which gives us detail view of Scheme names and their Net Asset Values in relation to Launch Dates.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 249 entries, 0 to 248
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Scheme NAV Name       249 non-null   object
1   Net Asset Value       249 non-null   float64
2   Date                  249 non-null   datetime64[ns]
3   Rating                207 non-null   float64
4   Riskometer            249 non-null   object
5   Launch Date           26 non-null    object
dtypes: datetime64[ns](1), float64(2), object(3)
memory usage: 11.8+ KB
```

Figure 1.5: Schemes Info

We got statistical parameters like count, mean, minimum value, maximum value, 25 percentile, 50 percentile, 75 percentile, standard deviation using describe function.

|       | Net Asset Value | Date                          | Rating     |
|-------|-----------------|-------------------------------|------------|
| count | 249.000000      | 249                           | 207.000000 |
| mean  | 478.749312      | 2024-10-26 17:03:36.867469824 | 2.922705   |
| min   | 0.000000        | 2015-06-08 00:00:00           | 1.000000   |
| 25%   | 11.324300       | 2025-06-27 00:00:00           | 2.000000   |
| 50%   | 15.108500       | 2025-11-07 00:00:00           | 3.000000   |
| 75%   | 33.928400       | 2025-11-07 00:00:00           | 4.000000   |
| max   | 15097.536000    | 2025-11-07 00:00:00           | 5.000000   |
| std   | 1750.017810     | NaN                           | 1.094477   |

Figure 1.6: Schemes Describe

### 1.3.1 SBI\_Fund\_NAV

This dataset contains 1186 daily NAV records of an SBI mutual fund scheme. It includes important fields such as Date, NAV value, daily Change.

|      | Date       | NAV     | Change | Change(%) |
|------|------------|---------|--------|-----------|
| 827  | 2022-06-01 | 27.1111 | 0.00   | 0.0000    |
| 87   | 2025-07-01 | 34.2744 | 0.03   | 0.0010    |
| 1143 | 2021-02-04 | 25.9518 | 0.01   | 0.0005    |
| 1185 | 2020-12-04 | 25.9998 | 0.04   | 0.0015    |
| 880  | 2022-03-10 | 27.1270 | 0.01   | 0.0005    |

Figure 1.7: SBI Sample

The dataset has four columns where Date is stored as a datetime field and the remaining values are numeric. This table gives a clear view about data size, non-null counts and data types.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1186 entries, 0 to 1185
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Date        1186 non-null   datetime64[ns]
1   NAV         1186 non-null   float64
2   Change      1183 non-null   float64
3   Change(%)   1175 non-null   float64
dtypes: datetime64[ns](1), float64(3)
memory usage: 37.2 KB

```

Figure 1.8: SBI Info

Using the describe function, we get statistical values such as count, mean, minimum, maximum, quartiles and standard deviation for NAV, Change.

|       | Date                          | NAV         | Change      | Change(%)   |
|-------|-------------------------------|-------------|-------------|-------------|
| count | 1186                          | 1186.000000 | 1183.000000 | 1175.000000 |
| mean  | 2023-05-26 07:26:48.768971264 | 29.453570   | 0.007768    | 0.000259    |
| min   | 2020-12-04 00:00:00           | 25.876400   | -0.120000   | -0.004400   |
| 25%   | 2022-02-24 06:00:00           | 27.073325   | 0.000000    | 0.000000    |
| 50%   | 2023-05-31 12:00:00           | 29.003150   | 0.010000    | 0.000200    |
| 75%   | 2024-08-25 06:00:00           | 31.696050   | 0.010000    | 0.000500    |
| max   | 2025-11-07 00:00:00           | 35.038600   | 0.120000    | 0.003800    |
| std   | NaN                           | 2.739042    | 0.015881    | 0.000528    |

Figure 1.9: SBI Describe

### 1.3.2 AXIS\_Fund\_NAV

This dataset contains 223 entries of NAV values for an Axis mutual fund scheme. It includes two fields: Net Asset Value and Date, which help in tracking how the fund's price changed over time.

|     | Net Asset Value | Date       |
|-----|-----------------|------------|
| 127 | 33.8034         | 2025-06-19 |
| 2   | 32.0357         | 2024-12-13 |
| 161 | 34.0901         | 2025-08-06 |
| 113 | 33.7386         | 2025-05-31 |
| 118 | 33.8710         | 2025-06-06 |

Figure 1.10: Axis Sample

The dataset has two columns where Net Asset Value is a float value and Date is stored as datetime. The info table shows the number of rows, non-null counts and data types.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 223 entries, 0 to 222
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Net Asset Value  223 non-null   float64
1   Date             223 non-null   datetime64[ns]
dtypes: datetime64[ns](1), float64(1)
memory usage: 3.6 KB
```

Figure 1.11: Axis Info

Using the describe function, we get statistical values like mean, minimum, maximum, quartiles and

standard deviation for the NAV field. This gives a good idea about the range and variation of the fund price over this period.

|       | Net Asset Value | Date                          |
|-------|-----------------|-------------------------------|
| count | 223.000000      | 223                           |
| mean  | 33.418824       | 2025-05-25 06:40:21.524663552 |
| min   | 32.028500       | 2024-12-11 00:00:00           |
| 25%   | 32.515500       | 2025-03-03 12:00:00           |
| 50%   | 33.736300       | 2025-05-29 00:00:00           |
| 75%   | 34.103900       | 2025-08-13 12:00:00           |
| max   | 34.668500       | 2025-11-07 00:00:00           |
| std   | 0.836388        | NaN                           |

Figure 1.12: Axis Describe

### 1.3.3 HDFC\_Fund\_NAV

This dataset contains 972 entries of NAV values for an HDFC mutual fund scheme. It includes two columns: Net Asset Value (NAV) and Date. These values help to understand the daily movement and historical performance of the fund.

|     | Net Asset Value | Date       |
|-----|-----------------|------------|
| 689 | 10.3344         | 2024-09-09 |
| 617 | 10.3602         | 2024-05-27 |
| 631 | 10.3217         | 2024-06-14 |
| 676 | 10.3294         | 2024-08-21 |
| 753 | 10.3074         | 2024-12-13 |

Figure 1.13: HDFC Sample



The dataset has two fields where Net Asset Value is stored as float and Date is stored as datetime. The info table shows the row count, non-null values and data types used in the dataset.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 972 entries, 0 to 971
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Net Asset Value  972 non-null    float64
1   Date             972 non-null    datetime64[ns]
dtypes: datetime64[ns](1), float64(1)
memory usage: 15.3 KB
```

Figure 1.14: HDFC Info

Using the describe function, we get statistics such as mean, minimum, maximum, quartiles and standard deviation for the NAV column. This gives a good picture about the NAV range and price variation throughout the time period.

|       | Net Asset Value | Date                          |
|-------|-----------------|-------------------------------|
| count | 972.000000      | 972                           |
| mean  | 10.319963       | 2023-11-05 14:05:55.555555840 |
| min   | 10.224800       | 2021-11-08 00:00:00           |
| 25%   | 10.309800       | 2022-11-06 06:00:00           |
| 50%   | 10.320200       | 2023-11-02 12:00:00           |
| 75%   | 10.331925       | 2024-11-06 06:00:00           |
| max   | 10.423400       | 2025-11-07 00:00:00           |
| std   | 0.021784        | NaN                           |

Figure 1.15: HDFC Describe

# Chapter 2. Data Cleaning

## 2.1 Data Preprocessing

In the data pre-processing section, we cleaned the data before doing any analysis on it. It is a very crucial step to convert raw, inconsistent data into some meaningful form. Inconsistencies are removed so that we can further explore the data properly. This enhances the quality and accuracy of the dataset. In the missing values in the dataset that we downloaded, in the place of null values, there were dashes in the excel file, which led to a great problem while converting the data type, dash is considered as string and we were unaware and hence we needed to remove dashes from the dataset.

In the raw dataset, the date columns were initially stored in pandas as object (string) type, which is not suitable for performing time-based operations. Therefore, we converted this into proper date-time format, to time series data. Because of this, we are easily able to extract month, date and year separately.

```
schemes['Launch Date'] = pd.to_datetime(schemes['Launch Date'], errors='coerce')
main_funds['Launch Date'] = pd.to_datetime(main_funds['Launch Date'], errors='coerce')
```

Figure 2.1: Datetime Describe

We used strip function to remove extra space in Schemes NAV Name so that all the scheme names become uniformly formatted and suitable for accurate comparison.

```
schemes['Scheme NAV Name'] = schemes['Scheme NAV Name'].apply(lambda x: x.strip())
main_funds['Scheme Name'] = main_funds['Scheme Name'].apply(lambda x: x.strip())
```

Figure 2.2: Pre-Processing/Strip

After converting them to strings, each fund name was processed using string splitting operations, where the text before the hyphen "-" was extracted. This helped isolate the primary or base fund name by removing trailing words such as plan types or options. This helps in proper formatting. We have

```
schemes['Fund Name'] = schemes['Scheme NAV Name'].astype(str).apply(lambda x: x.split("-")[0])
main_funds['Scheme Name'] = main_funds['Scheme Name'].astype(str).apply(lambda x: x.split("-")[0])
schemes['Fund Name'] = schemes['Fund Name'].apply(lambda x: x.strip())
```

Figure 2.3: Pre-Processing/Split

dropped a column Net Assets (Cr) because it was exactly similar to other column and so there was redundancy which is necessary to be removed.

```
main_funds.drop(columns=['Net Assets (Cr)'],inplace = True)
main_funds.drop(['Plan','Category Name'],axis=1,inplace=True)
```

Figure 2.4: Pre-Processing/Drop

Now, here we have percentage values as fraction in the original dataset like in 1 Week, 1 Month and similar columns and hence we have multiplied them with 100, to get them in percentage values out of 100.

```
columns = ['1W', '1M', '3M', '6M', 'YTD', '1Y', '2Y', '3Y', '5Y', '10Y', 'Expense Ratio (%)']
main_funds[columns] = main_funds[columns] * 100
display(main_funds.head())
```

Figure 2.5: Pre-processing by multiplying with 100

## 2.2 Missing data analysis

Missing data in a dataset can significantly affect the quality and reliability of the analysis. Identifying and addressing missing values is one of the key steps in data pre processing. In this section, we present how we analyzed and handled missing data.

The code: `data.isnull().sum()` checks for missing values in each column of the data set, returning the total number of null values per column. The results from this function revealed that certain columns, such as such as YTM, 1M, YTD, 1Y, 3Y, 10Y and Launch date had many missing values among other columns such as Fund name and Scheme NAV Name.

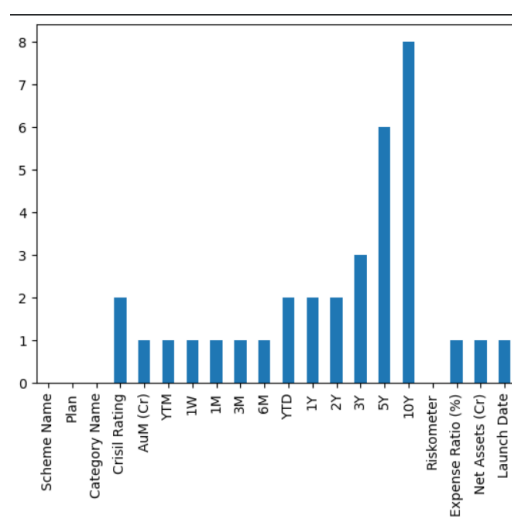


Figure 2.6: No. of Missing values

**Bar Plot of Missing Values:** The bar plot visualizes the missing values across the data set. It allows us to quickly identify columns with a high proportion of missing data. For example, columns with large bars indicate a significant percentage of missing values, which can be addressed by techniques such as imputation or removal based on the modeling strategy, columns, like Riskometer and Launch Date have high number of Missing Values.

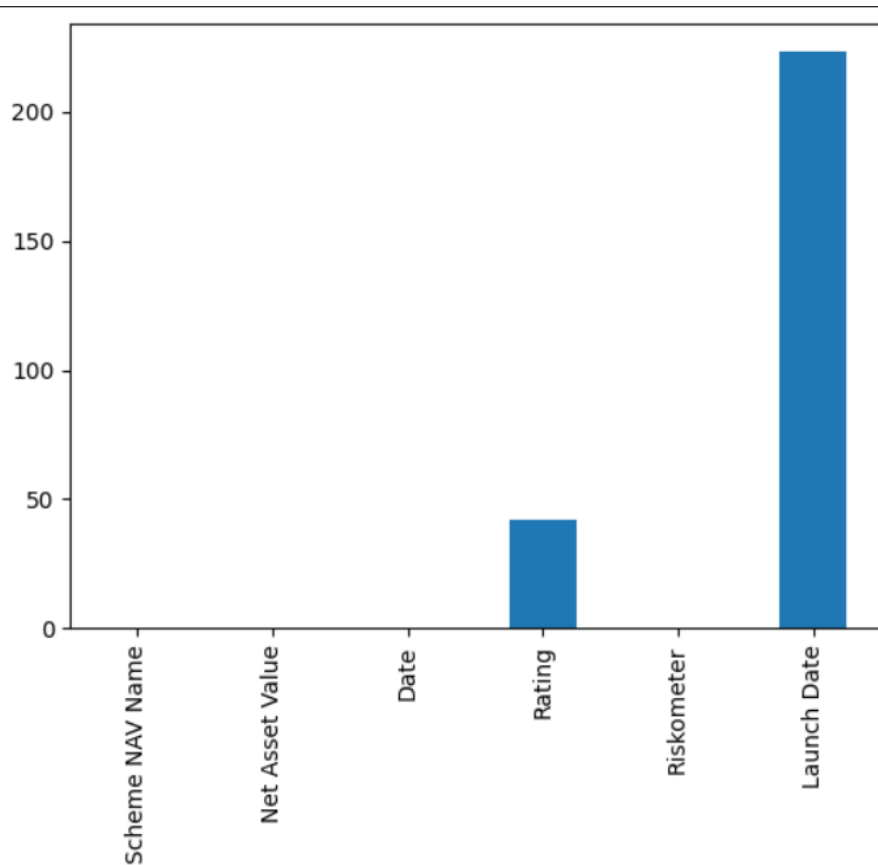


Figure 2.7: No. of Missing values

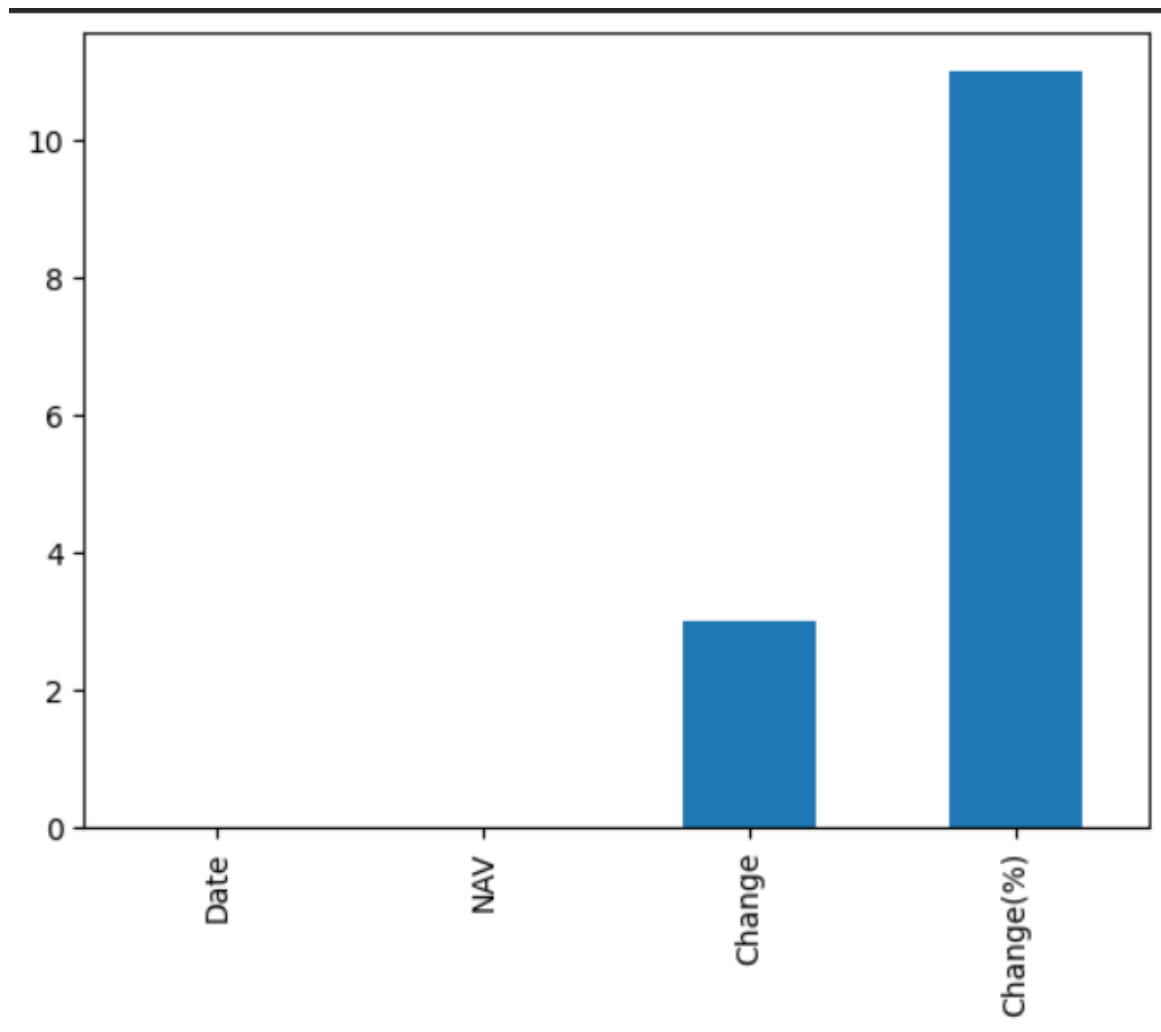


Figure 2.8: Dataset Describe

The heatmap shows us where missing values occur in the dataset. Black areas indicate that the data is complete, while the light yellow regions show columns that contain missing values. Features such as Rating, Riskometer, Expense Ratio, Net Assets, and Launch Date show noticeable patches of missing data. This visualization helps identify which columns require cleaning or imputation. It gives insight into patterns of missingness. It becomes easier to decide the right pre-processing steps for reliable analysis.

Below is the plot of Heatmap for visualizing the null Values in main\_funds database.

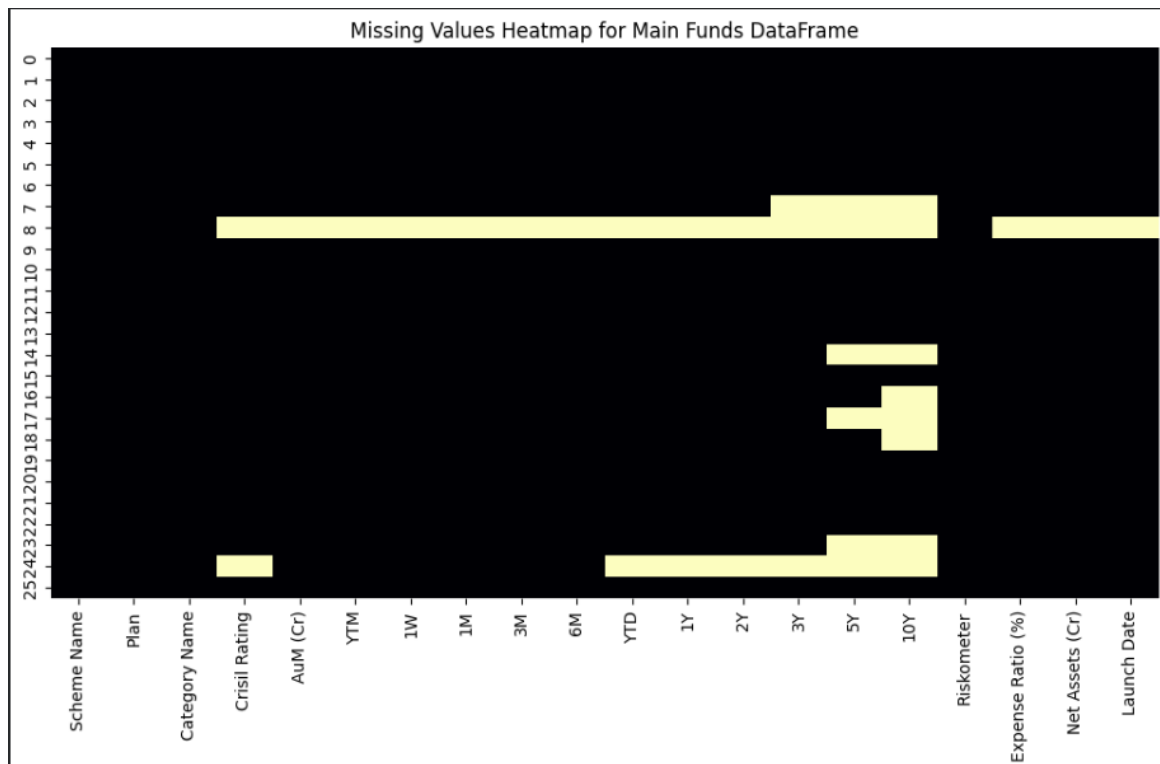


Figure 2.9: Heat Map of the Main Funds Dataset

Below is the plot of Heatmap for visualizing the Null Values in Schemes.

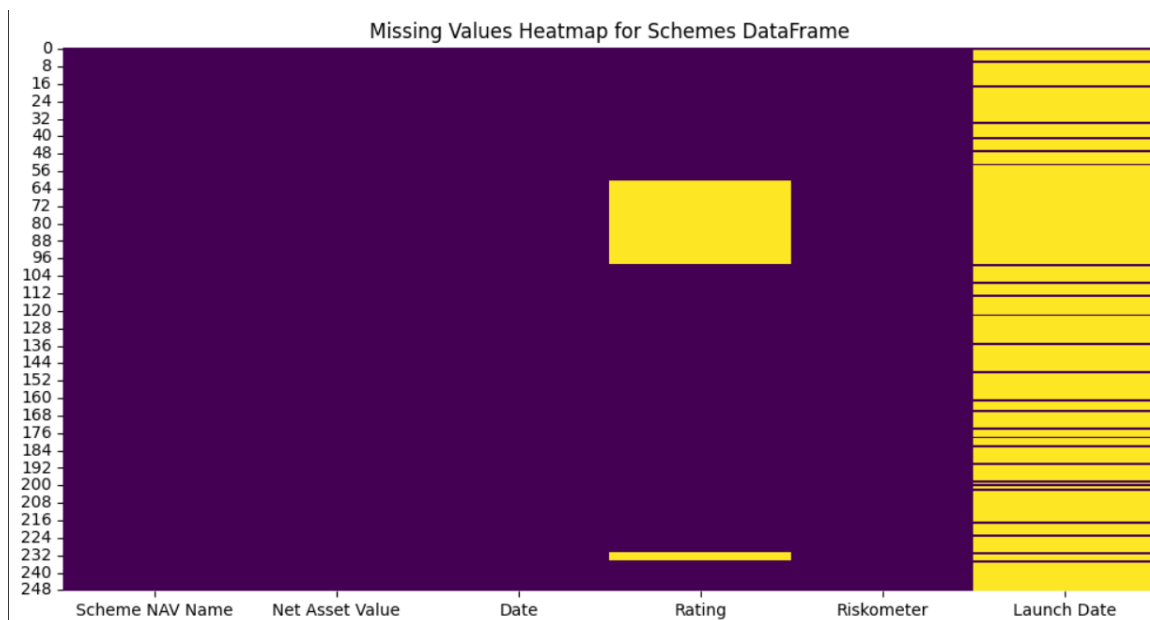


Figure 2.10: Heat Map of the Schemes Dataset

## 2.3 Imputation

After identifying and analyzing the missing values in the dataset, the next step in the data-cleaning process is imputation. Imputation means replacing missing entries with estimated or calculated values to ensure the dataset remains complete and usable. We used three methods for imputation:

1. Dropping rows with excessive missing values, where the data was too incomplete to recover meaningfully.
2. Forward fill (ffill)
3. Mean value imputation, where numerical features with scattered missing values were filled using their respective mean values.

### 2.3.1 Forward Fill(ffill)

Forward Fill (ffill): After applying backfill, the forward fill method propagates the last valid observation forward to fill any remaining missing values. This method ensures that no missing values remain in the data set. This provides a complete data set for further analysis and modeling. The effectiveness of these imputation techniques was confirmed using a Missing Value Bar Plot, which shows that after imputation, no missing values remained in the dataset, making it ready for analysis.

```
schemes['Launch Date']=schemes['Launch Date'].ffill()  
main_funds['Launch Date']=main_funds['Launch Date'].ffill()
```

Figure 2.11: Forward Fill

### 2.3.2 Complete Case Analysis(drop)

Complete Case Analysis is one of the simplest approaches to handling missing data. In this method, any row that contains many missing values is removed entirely from the dataset.

```
main_funds.drop(index=8, inplace=True)
```

Figure 2.12: Imputation Drop

### 2.3.3 Mean Imputation

In this technique, we just fill the missing values with the average (mean) of the non-missing observation values from the same feature. This is a common technique.

```

main_funds.loc[main_funds['Crisil Rating'].isnull(), 'Crisil Rating'] = np.ceil(main_funds['Crisil Rating'].mean())

main_funds.loc[main_funds['YTD'].isnull(), 'YTD'] = main_funds['YTD'].mean()

main_funds.loc[main_funds['1Y'].isnull(), '1Y'] = main_funds['1Y'].mean()

main_funds.loc[main_funds['2Y'].isnull(), '2Y'] = main_funds['2Y'].mean()

main_funds.loc[main_funds['3Y'].isnull(), '3Y'] = main_funds['3Y'].mean()

main_funds.loc[main_funds['5Y'].isnull(), '5Y'] = main_funds['5Y'].mean()

main_funds.loc[main_funds['10Y'].isnull(), '10Y'] = main_funds['10Y'].mean()

sbi_data.loc[sbi_data['Change'].isnull(), 'Change'] = sbi_data['Change'].mean()

sbi_data.loc[sbi_data['Change(%)'].isnull(), 'Change(%)'] = sbi_data['Change(%)'].mean()

schemes.loc[schemes['Rating'].isnull(), 'Rating'] = np.ceil(schemes['Rating'].mean())

```

Figure 2.13: Imputation of Mean

## 2.4 Outlier detection

Outliers are data points that significantly deviate from the normal range or pattern of the dataset. They are generally seen due to data entry mistakes or unusual events, or natural variability within the data. Outliers can have a strong impact on statistical calculations, such as the mean and standard deviation. They may also distort visualizations or affect the performance of analytical models. There are many techniques to identify outliers such as box-plots and scatter plots. We have used Inter-Quartile Range (IQR) as the data is right skewed.

This is the box plot of Net Asset Value from schemes dataset, and we can see the outliers in this plot.

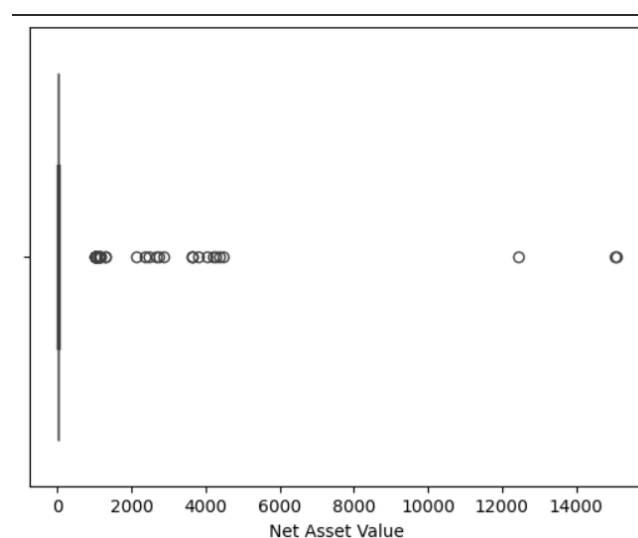


Figure 2.14: Outlier NAV



This is a histogram for Net Asset Value, which clearly shows us that this is a right skewed data, and there are few outliers (extreme right data points) near 14000. We need to work upon this. As this is a right skewed data, IQR method is useful for detecting and removing outliers.

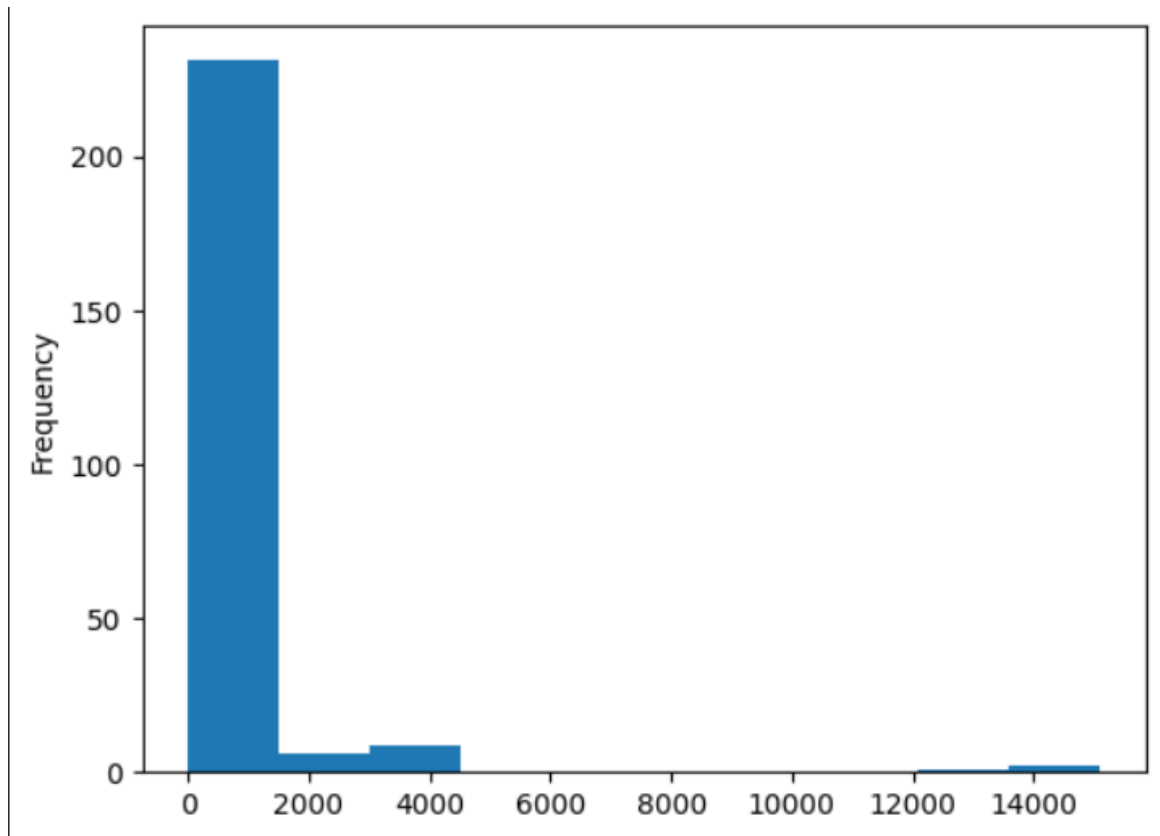


Figure 2.15: Outlier Before IQR

This is the box plot of 5Y returns from the main\_funds dataset, and we can see that there are numerous outliers.

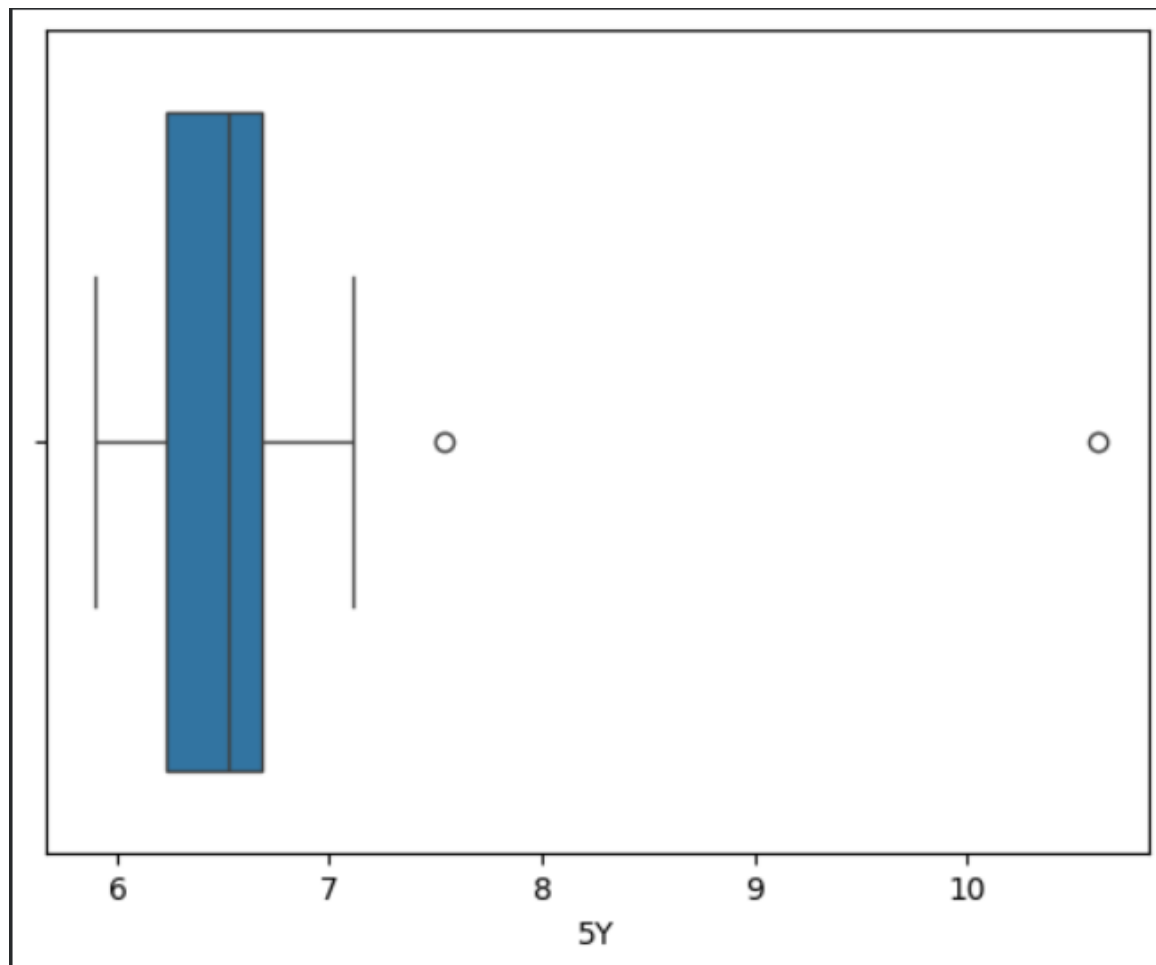


Figure 2.16: Outlier 5 Years

Isolation Forest is a good technique for detecting outliers across an entire dataset. Unlike traditional methods such as box plots, which identify outliers one column at a time, Isolation Forest analyzes all numerical features simultaneously in the data. It can detect multi-dimensional outliers.

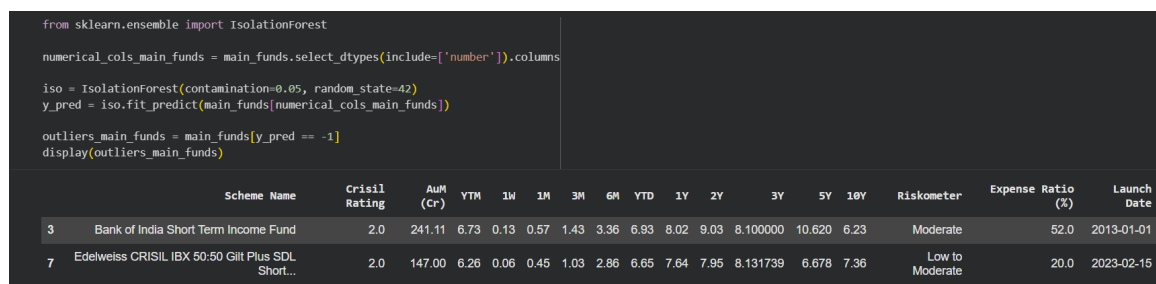


Figure 2.17: Outlier Detection Isolation

```
Q1 = schemes_copy['Net Asset Value'].quantile(0.25)
Q3 = schemes_copy['Net Asset Value'].quantile(0.75)
IQR = Q3 - Q1
upper_bound = Q3 + 1.5 * IQR

schemes_copy['Net Asset Value'] = np.where(schemes_copy['Net Asset Value'] > upper_bound, upper_bound, schemes_copy['Net Asset Value'])

sns.boxplot(x=schemes_copy['Net Asset Value'])
```

Figure 2.18: Outlier Imputation

The Inter-Quartile Range (IQR) method is a famous statistical method for detecting and removing outliers. It is based on the spread of the data. Thus, here for Net Asset Value Column in the schemes data set, we have used IQR method to remove the outliers and then we have plotted the box plot.

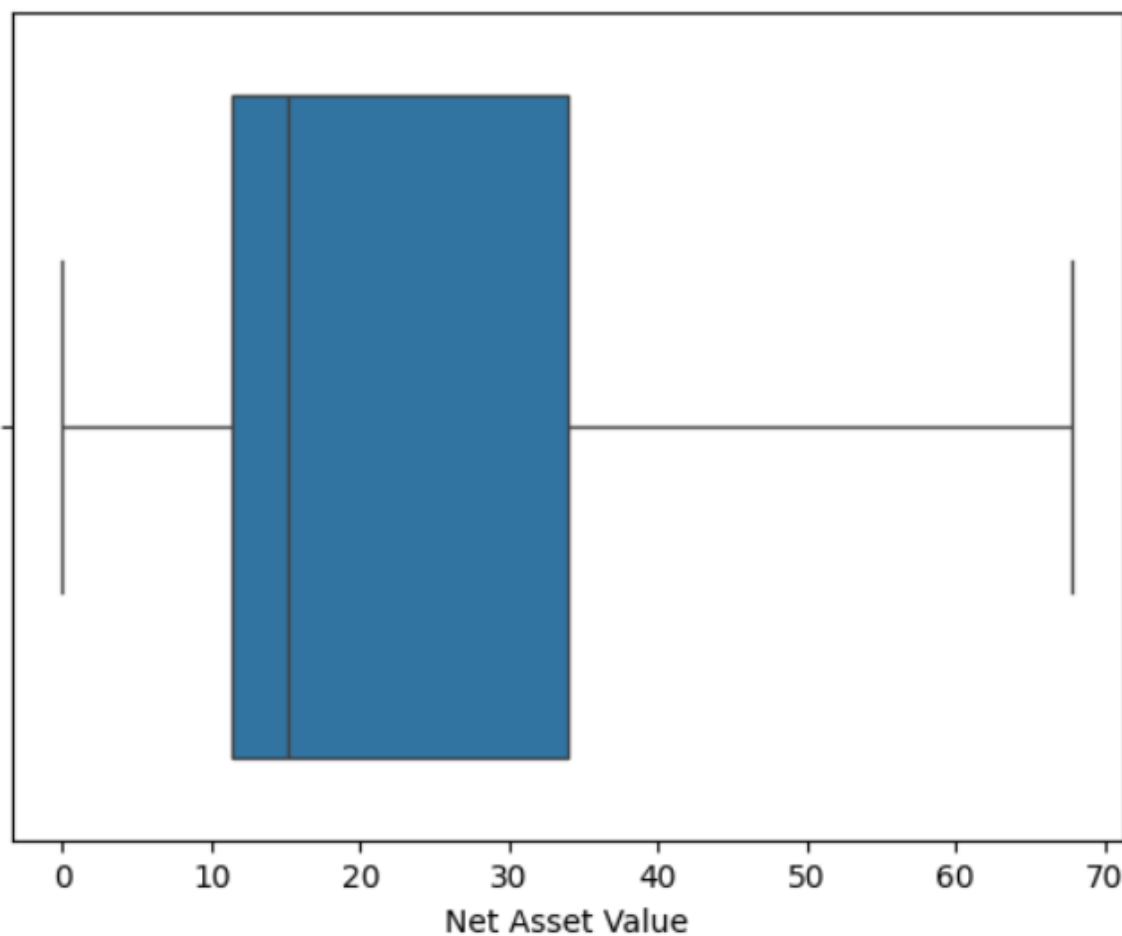


Figure 2.19: Box Plot of Net Asset Value

We have also plotted the Histogram for the Net Asset Value after applying the IQR method on it and we have removed the outliers from it, meaning those data points which are greater than the upper bound are removed. Initially it was right skewed data, but it became more representative and suitable for further analysis.

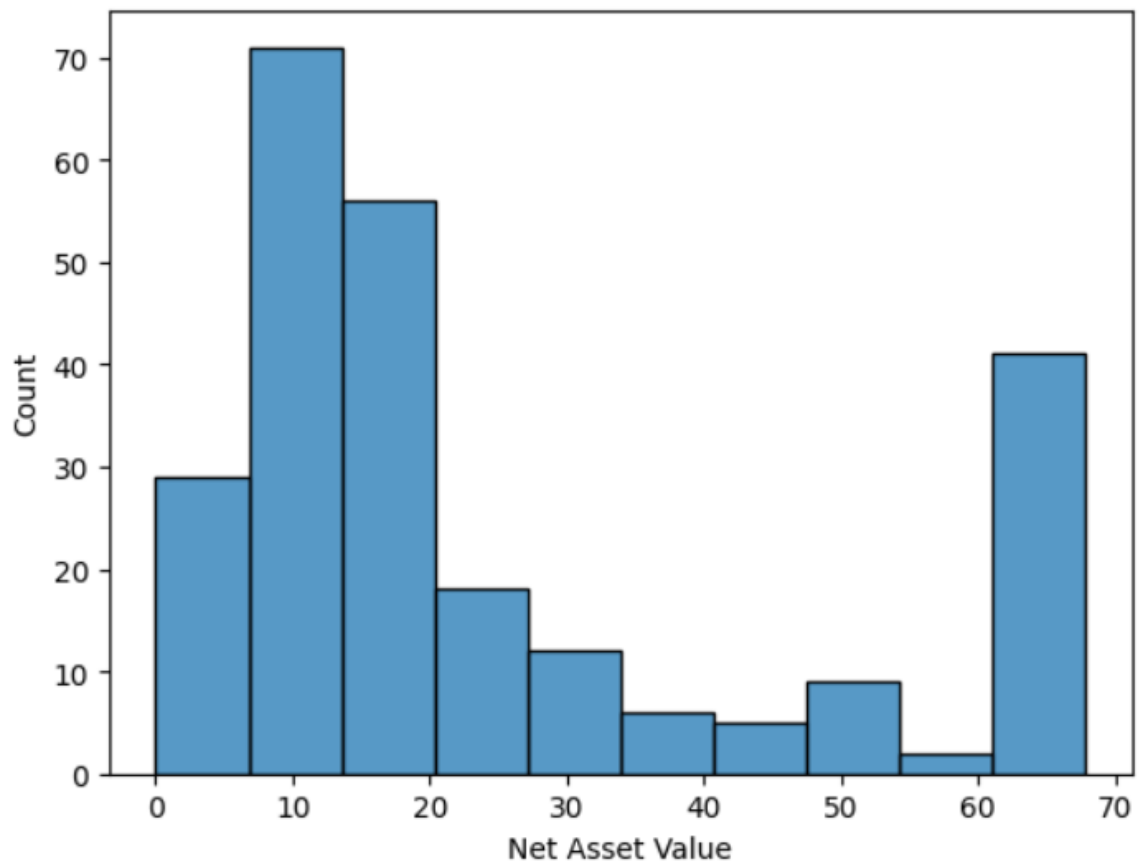


Figure 2.20: Net Asset Value (NAV)

# Chapter 3. Visualization

Visualization is an important step in exploratory data analysis as it helps to convert the data into understandable and visually pleasing insights. It helps in finding patterns and trends. We can easily interpret some complex datasets using visualization techniques such as: Histogram, count plot, Scatter Plot, Pie Chart.

## 3.1 Univariate analysis

Univariate analysis focuses on examining the distribution and characteristics of a single variable at a time. The objective is to understand its central tendency, spread, range, and overall behavior.

### 3.1.1 Distribution of funds returns(Main Funds)

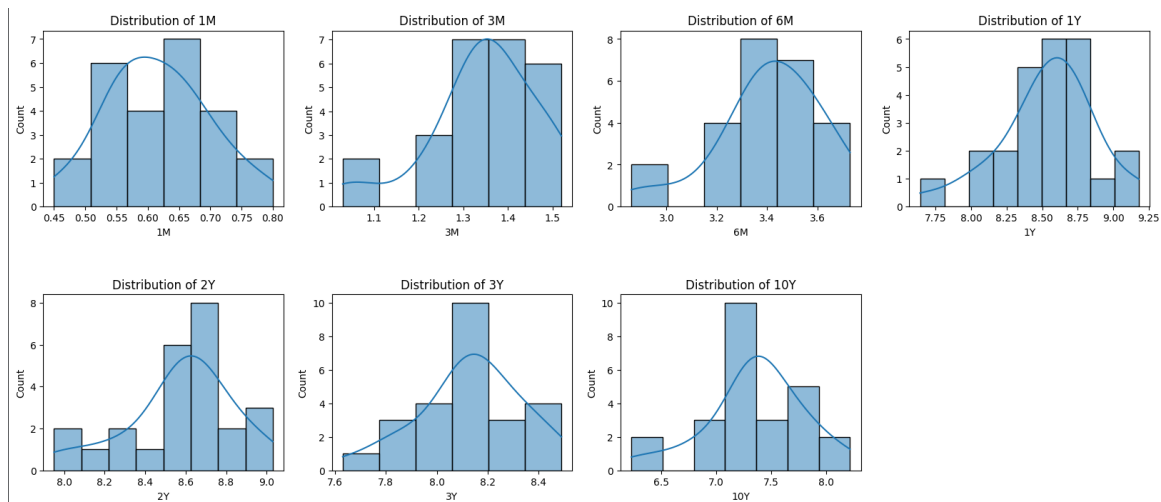


Figure 3.1: Series of Histogram showing distribution of fund returns across time periods

The above figure is a series of univariate histograms for key performance metrics of Short Term Debt Funds dataset, including 1M, 3M, 6M, 1Y, 2Y, 3Y and 10Y returns. We have combined histograms with kde plot (kernel density estimation curves). This plot help us to observe how each metric is distributed. 1 Month Return, 1 Year Return and 3 Year Return distribution shows a fairly symmetric spread, without any strong skewness.

3 Month Return distribution appears slightly right skewed showing higher positive returns.

6 Month Return and 2 Year returns distribution is roughly normal with minor right skewness.

### 3.1.2 Distribution of YTD

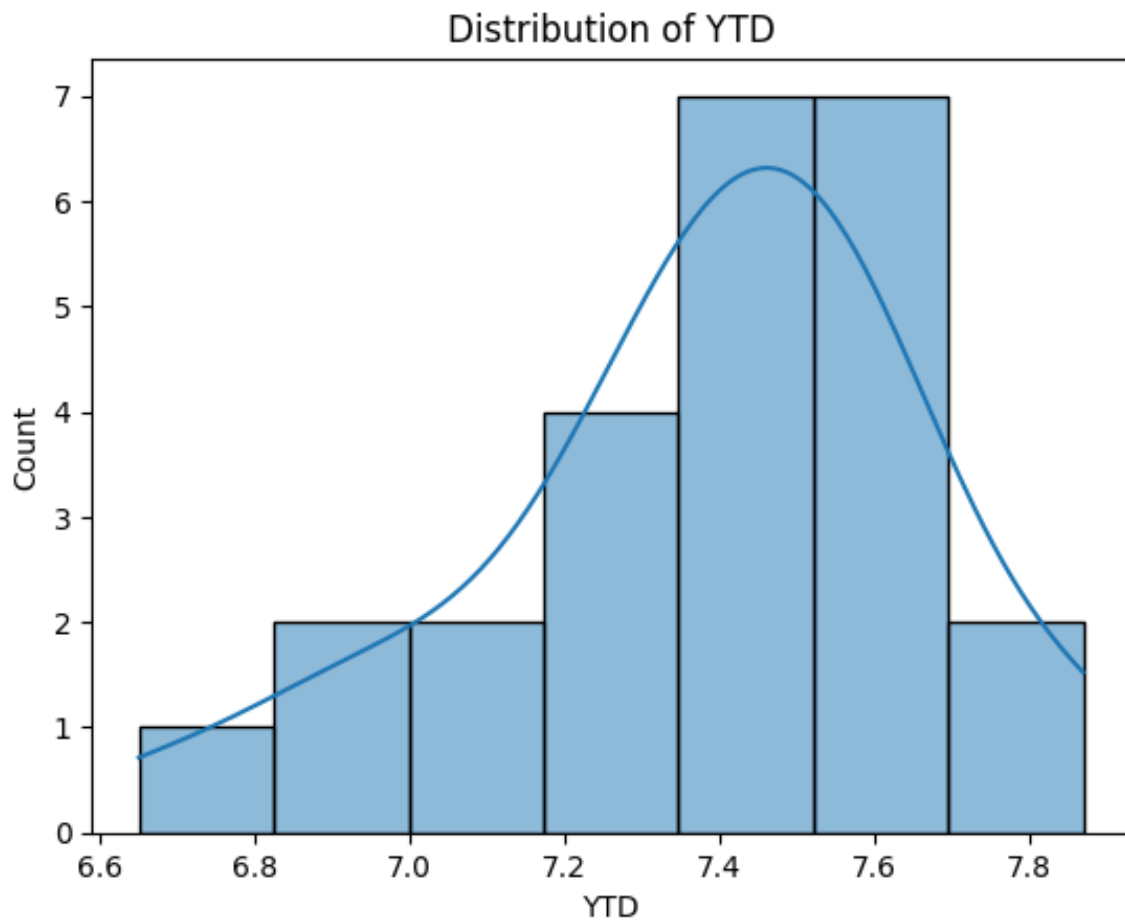


Figure 3.2: Series of Histogram showing distribution of fund returns across time periods

YTD (Year To Date):

It return measures the fund's performance from the beginning of the current calendar year up to the present date. It reflects how much the fund has gained or lost during this period, capturing short-term market movements and recent trends.

This graph shows the distribution of YTD returns across the 26 short-term debt funds. Most of the funds have YTD returns between 7.2% and 7.6%, indicating very low variation in performance. The highest bars and the KDE peak show that the typical YTD return lies around 7.4% to 7.6%. Only a few funds fall at the extremes, with lower YTD values around 6.6%–6.9% and higher values near 7.8%. The bell-shaped curve also indicates that the YTD returns follow a near-normal distribution, suggesting stable and consistent performance across the funds.

### 3.1.3 Distribution of YTM in Main Funds

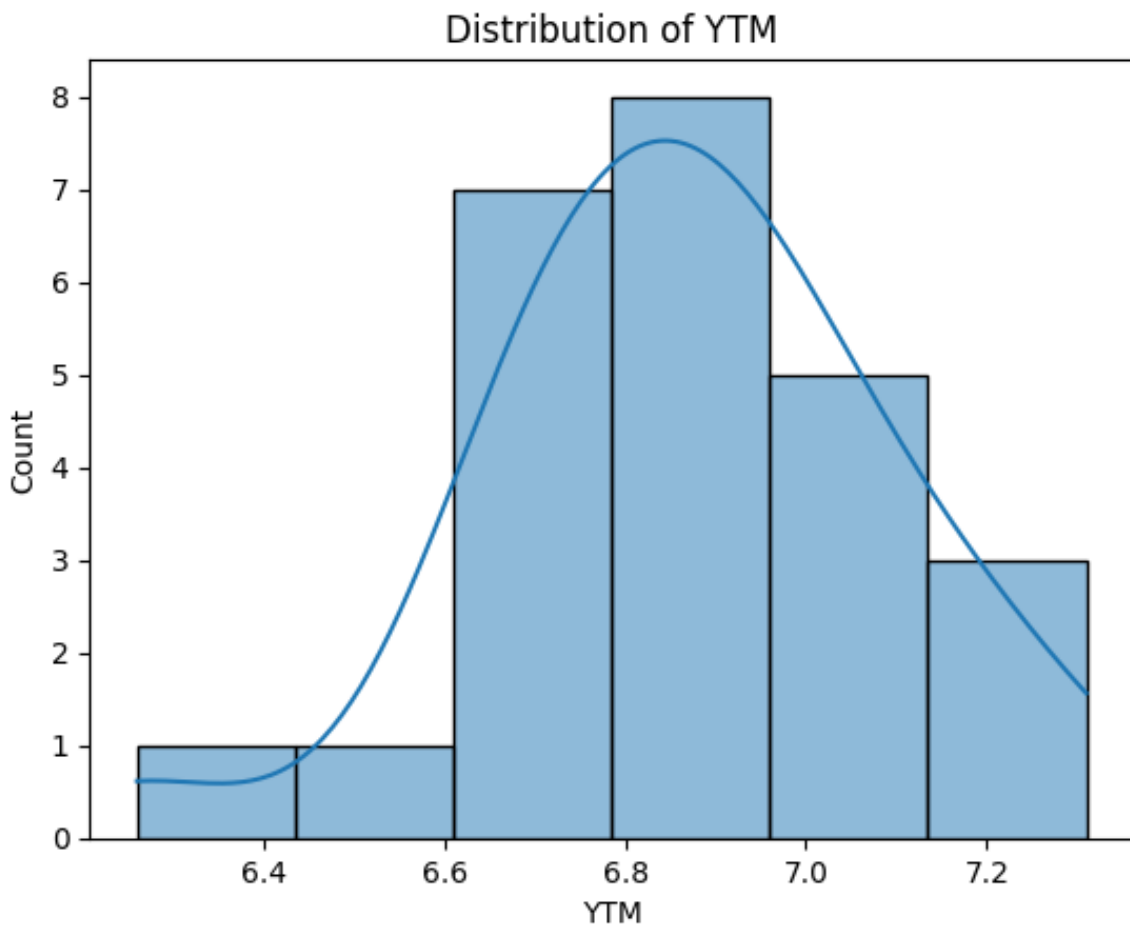


Figure 3.3: Series of Histogram showing distribution of fund returns across time periods

YTM (Yield To Maturity):

It is the total expected return an investor would earn on a bond or security if it is held until it matures. This graph shows the distribution of YTM across the 26 short-term debt funds. Distribution is nearly a normal distribution around a mean of approximately 6.9%. The majority offering yields between 6.8% and 7.0%. A slightly negative skew also reveals the presence of slightly low yields of around 6.3%. The upside part is capped near 7.2% which mainly describes the competitive nature of short duration debt market.

### 3.1.4 Distribution of Expense Ratio across Debt Funds

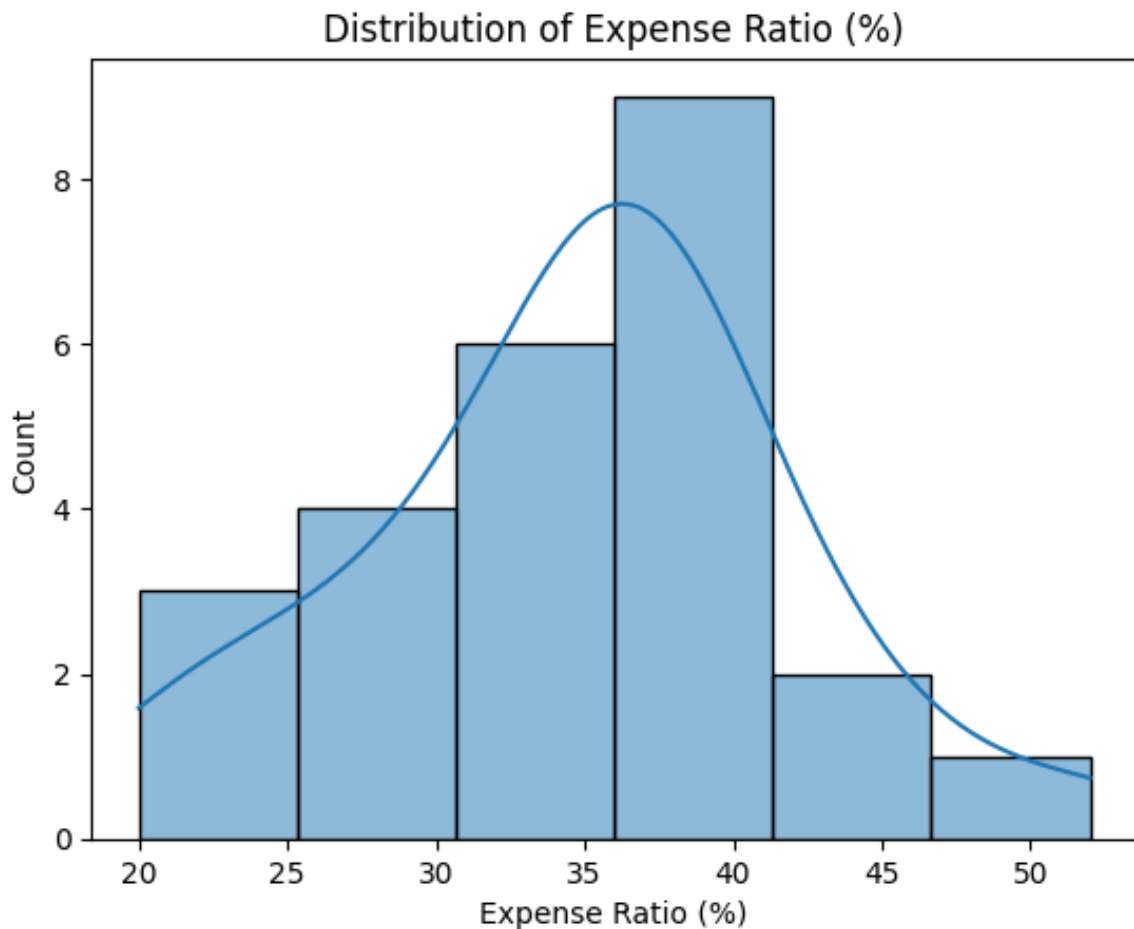


Figure 3.4: Distribution of Expense Ratios across 26 short-term debt funds

This graph shows the distribution of Expense Ratios across the 26 short-term debt funds. Most funds have an expense ratio between 30% and 42%, indicating that the majority follow a similar expense ratio. The highest bars and the KDE peak show that the typical expense ratio lies around 35% to 40%, which represents the most common operating cost range among these funds. A few funds fall at the lower end around 20%–25%, while very few reach the higher end near 50%, showing some variation but not extreme. The overall shape of the curve is bell-shaped, suggesting that expense ratios are following near to normal distribution.



### 3.1.5 Distribution of Fund Launch Years (Main funds)

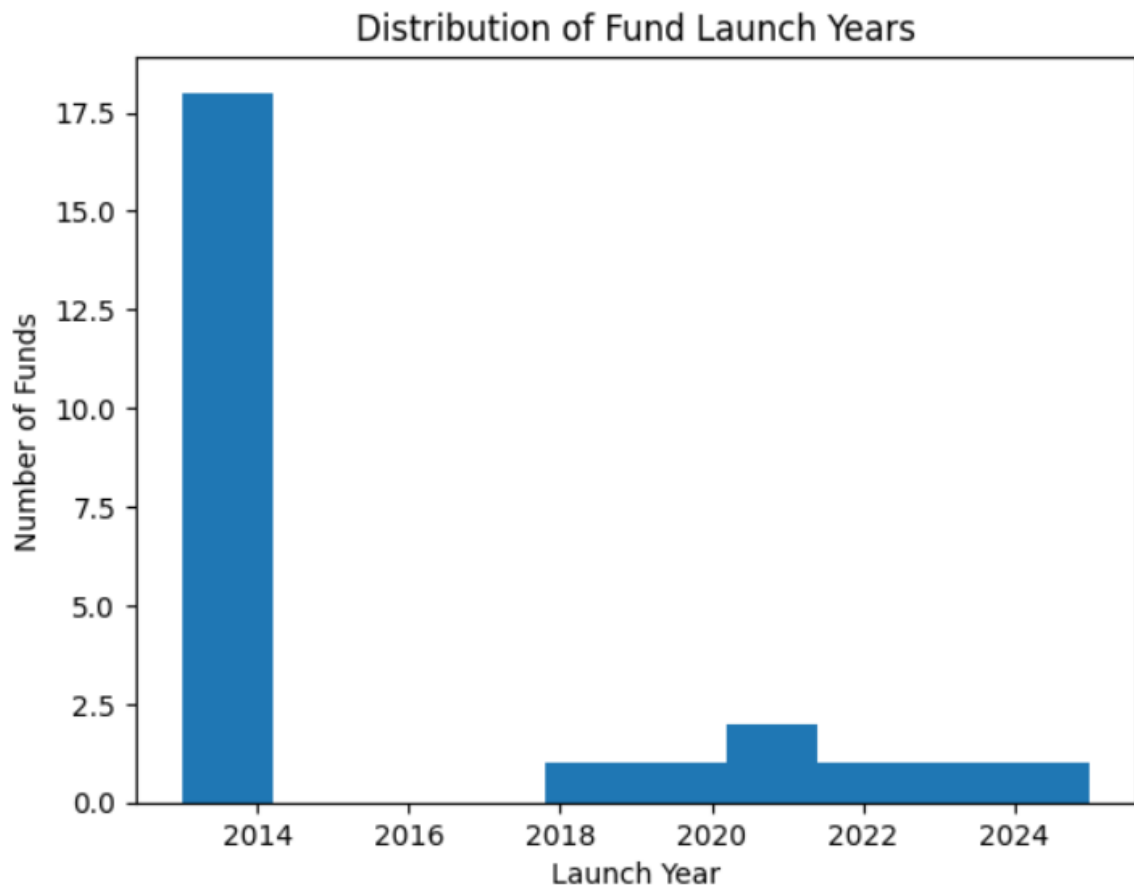


Figure 3.5: Bar plot of Fund Launch Years

It is a very interesting plot as the majority of the Short Duration Debt Funds in the dataset were launched around 2014. This means launching of funds were in a concentrated period.

Very few funds were launched between 2016 and 2018, showing a slowdown in new fund creation.

Many existing funds have been operating for several years, providing sufficient historical data.

### 3.1.6 Distribution of Scheme Ratings (Schemes)

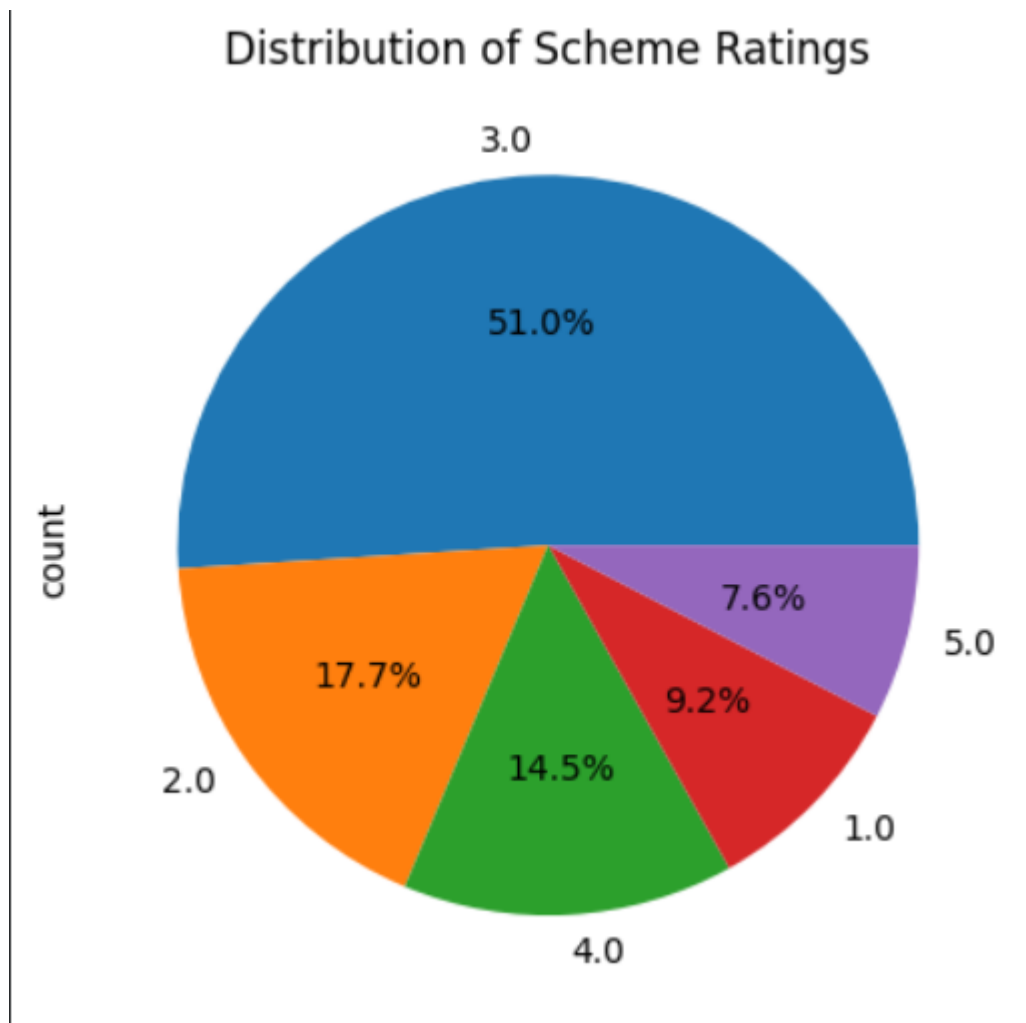


Figure 3.6: Pie Chart of Scheme Rating

The pie chart illustrates distribution of scheme ratings across the dataset. The ratings are from 1 to 5. This distribution suggests that most short-duration debt funds fall in the mid-range rating category. This indicates moderate credit quality and performance stability. There are very few extremely high rated or low rated funds.

### 3.1.7 Distribution of Scheme Riskometer (Schemes)

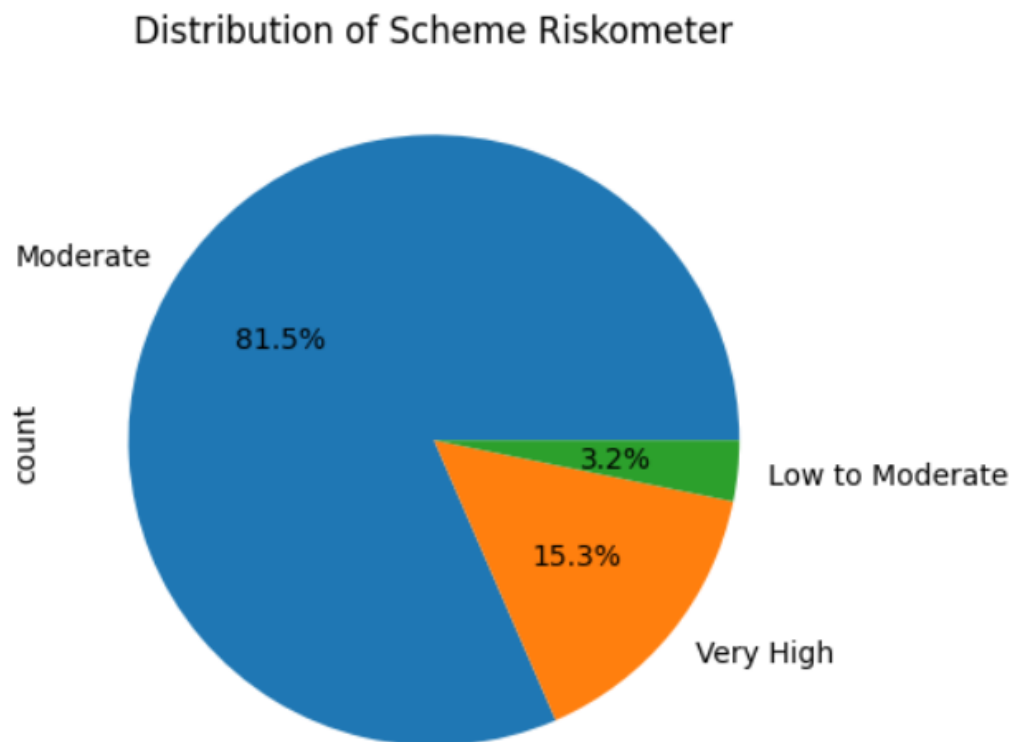


Figure 3.7: Pie Chart of Riskometer

The pie chart illustrates the distribution of funds based on their Riskometer classifications. This gives insight into the overall risk profile of the Short Duration Debt Fund category. A large majority of the schemes, approximately 81.5%, fall under the Moderate risk category. Most funds aim to balance safety with reasonable returns.

## 3.2 Multivariate analysis

Multivariate analysis focuses on relationships between two or more variables simultaneously, to get insights into datasets and patterns. For example, Scatter plots are a good way to plot relation between two or more variables.

### 3.2.1 Plot of YTM and YTD (Main Funds)

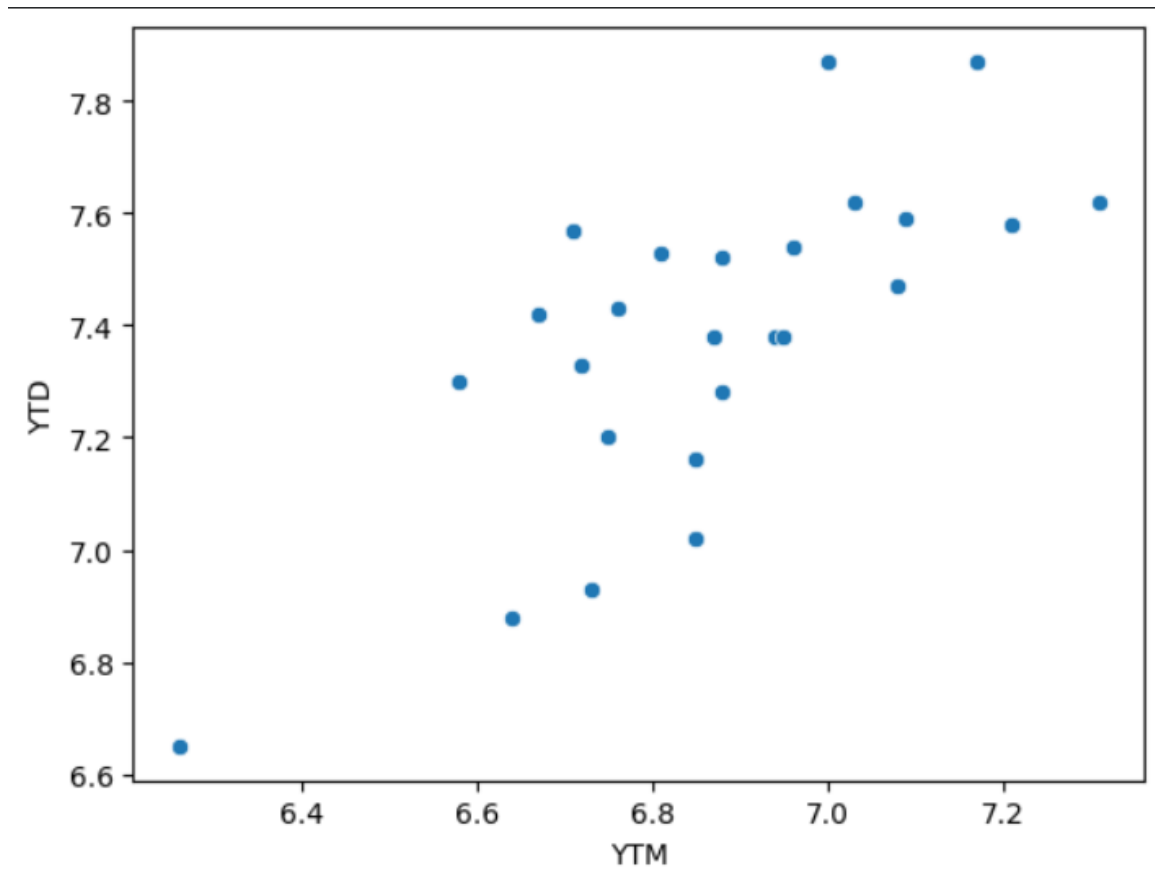


Figure 3.8: YTM vs YTD

As we have already discussed about the two terms earlier above in the report, here is the graph showing the relation between the two using the scatter plot.

The scatter plot shows the relationship between Yield to Maturity (YTM) and Year-to-Date (YTD) returns for the different short-duration debt funds. Each point represents one fund, and the overall pattern suggests a roughly increasing trend. This means that funds with a higher YTM tend to also show higher YTD returns, although the relationship is not perfectly straight.

In simple terms, when the expected yield of the underlying bonds (YTM) is higher, the actual performance so far in the year (YTD) also tends to be stronger. The points do not form a perfect line, but they show a general positive direction. This indicates a mild to moderate linear relationship. This suggests that YTM can provide some indication of recent performance, but other factors also influence YTD returns.

### 3.2.2 Visualizing Dendrogram (Main Funds)

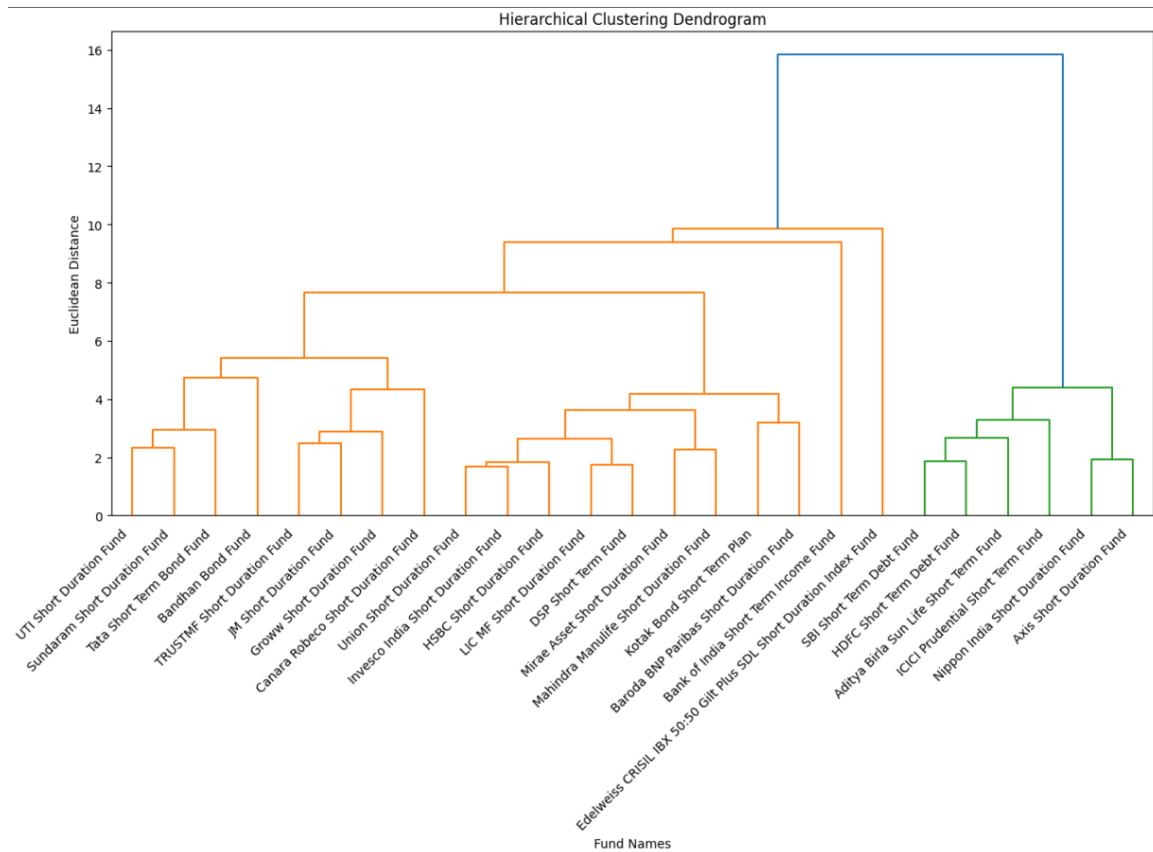


Figure 3.9: Dendrogram

The dendrogram provides a hierarchical clustering grouping columns with similar data patterns. Dendrogram is used to identify which funds behaves similarly in terms of numerical columns like returns, expense ratio and NAV values and also identify which are the outliers.

1. We can see two different clusters in graph: Orange Cluster(Left) and Green Cluster(Right). The dendrogram shows a primarily split at Euclidean distance of approximately 16.
2. The cluster on the right mainly consists of India's largest Asset Management Companies like SBI, HDFC, Axis. Their tight group suggests that these market leaders employ highly correlated strategies. So if an investor holds both HDFC and SBI short term debt funds, they are likely to achieve same benefits.
3. The cluster on the left contains a mix of mid-sized and small AMCs. This orange branching suggests these funds are more active in management styles with a wider spectrum of risk.
4. Several funds show extremely high similarity of approximately 2 unit of euclidean distance. So investor can substitute one fund with another with minimum impact on his portfolio.

### 3.2.3 Multi Line plot of Funds Returns Over time (Main Funds)



Figure 3.10: Multi-line plot

We have selected different time-period return columns (from 1 week to 10 years) and plots them for each mutual fund. For every fund (each row in the dataset), a line plot is drawn showing how its returns change across these durations.

The line chart illustrates how the returns of various Short Duration Debt Funds evolve across different investment horizons, ranging from 1 week to 10 years. In the short-term durations (1W, 1M, 3M, 6M), the returns show a steady upward movement, reflecting the typical behavior of short-duration debt instruments that accumulate returns gradually over time.

A significant rise is observed at the 1-year and 2-year marks, where most funds reach their highest return levels. This peak suggests that short-duration debt funds tend to perform best in the 1–2 year holding period.

Beyond this point, a noticeable dip occurs in the 3-year and 5-year returns, indicating that medium term performance is comparatively weaker.

Interestingly, at the 10-year horizon, the returns rise again for several funds.

This visualization effectively shows how return behavior varies across durations. It helps investors identify the optimal investment horizon for short-duration debt funds.

### 3.2.4 Name of Fund vs Net Asset Value Plot

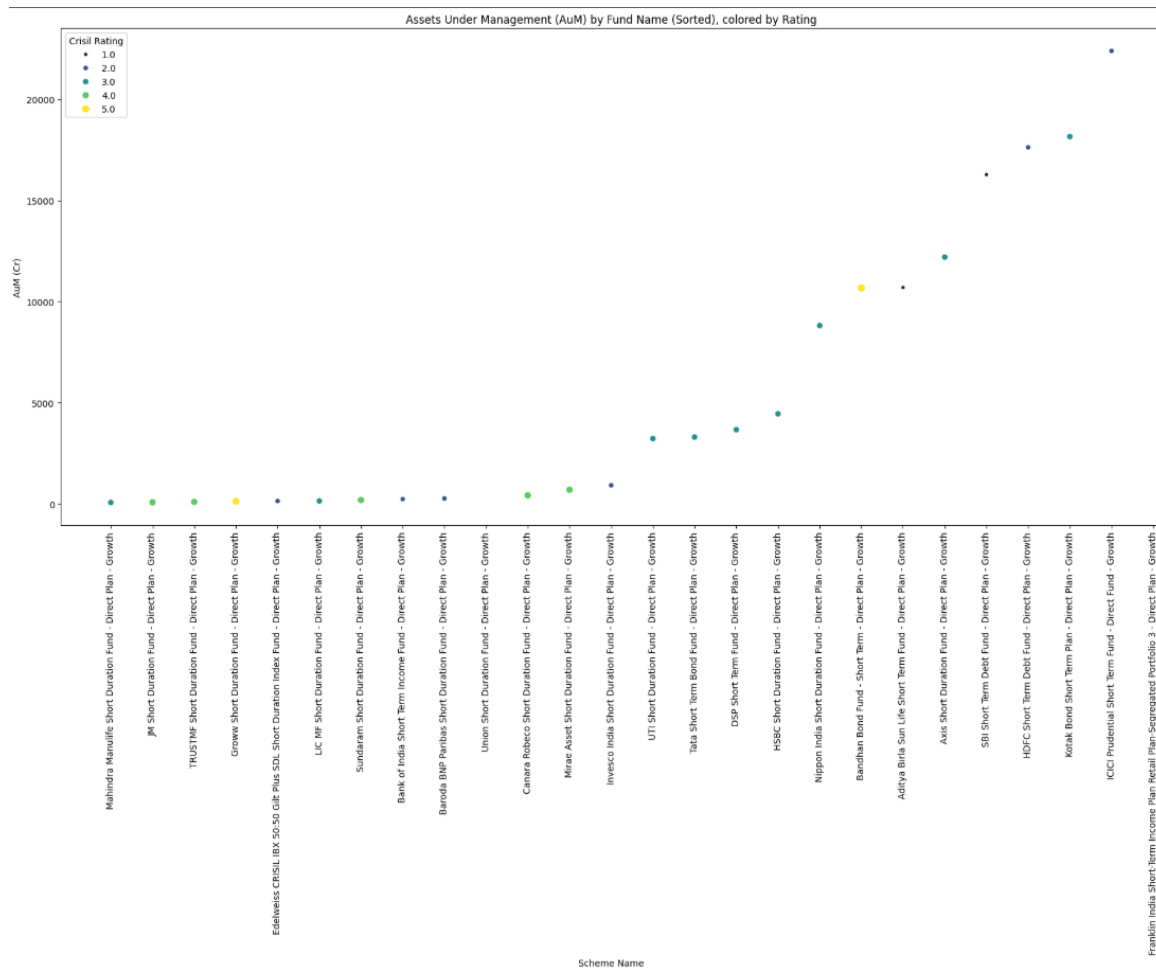


Figure 3.11: Name vs AuM

This scatter plot shows the relationship between different funds and their Assets Under Management (AuM in crore). Each point represents a fund, and the height of the point indicates the size of its AuM. The points are color-coded based on their CRISIL ratings, allowing easy comparison across different risk categories. The plot clearly highlights that some funds manage extremely high AuM while many others manage smaller amounts, creating a noticeable imbalance among fund sizes.

#### Key Observations

**Power Law Pattern:** Large fund houses such as HDFC, SBI, and ICICI hold significantly higher AuM compared to smaller companies. This reflects a power-law distribution, where a few dominant players control most of the assets while the majority hold comparatively small amounts.

**No Link Between AuM and Rating:** The plot also shows no direct relationship between high AuM and CRISIL rating. Some funds with very large AuM have only moderate ratings, while several smaller funds have stronger ratings. This indicates that fund size does not necessarily represent credit quality or risk level.

### 3.2.5 Distribution of Scheme Ratings (Schemes)

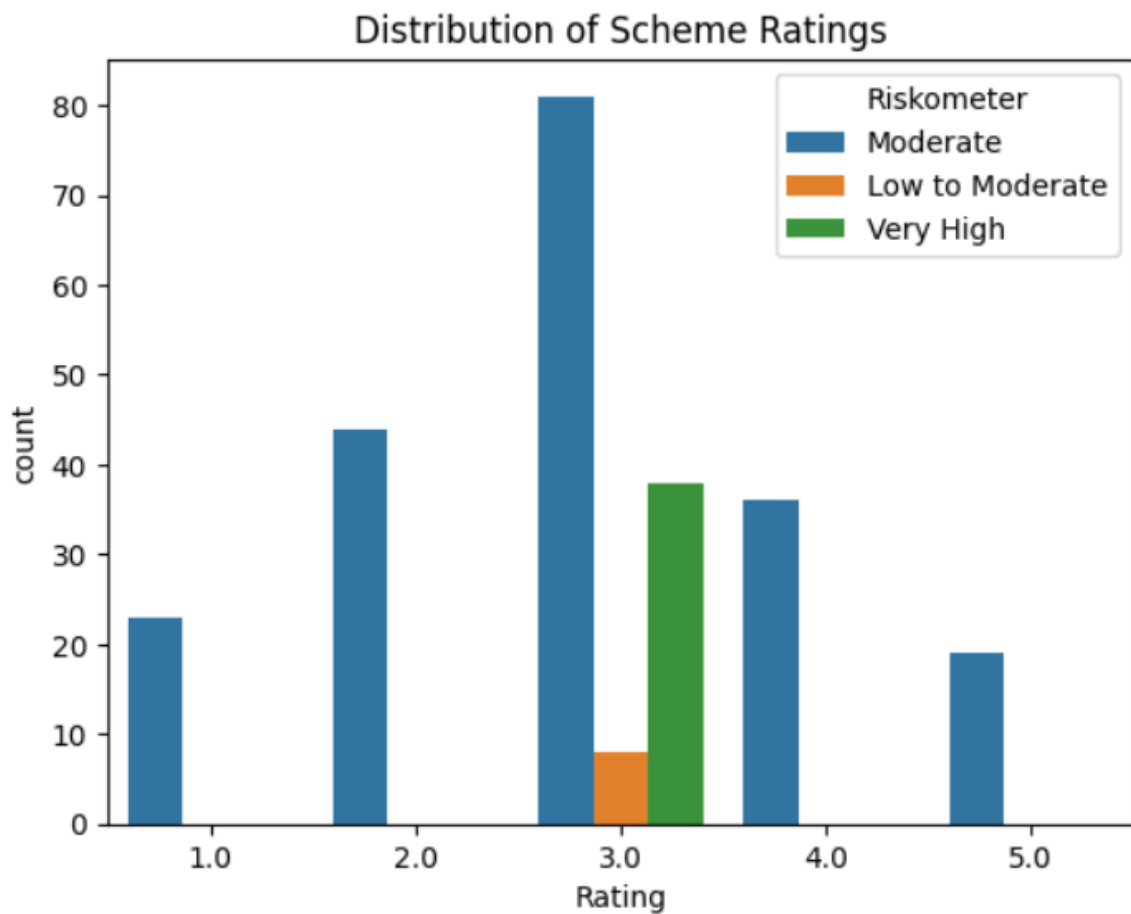


Figure 3.12: Distribution of Scheme Ratings: Count Plot

The above visualization is a count plot that displays how many schemes fall under each rating category. While doing so, we also grouped them by their Riskometer classifications. The plot shows that the rating of 3 has the highest number of schemes, with most of them classified under the Moderate risk category, indicating a strong concentration of average-rated and moderate-risk funds. Ratings 1 and 2 also have noticeable representation, again dominated by Moderate-risk schemes. This highlights the conservative nature of most short-duration debt funds.



### 3.2.6 Distribution of Net Asset Value Grouped by Riskometer

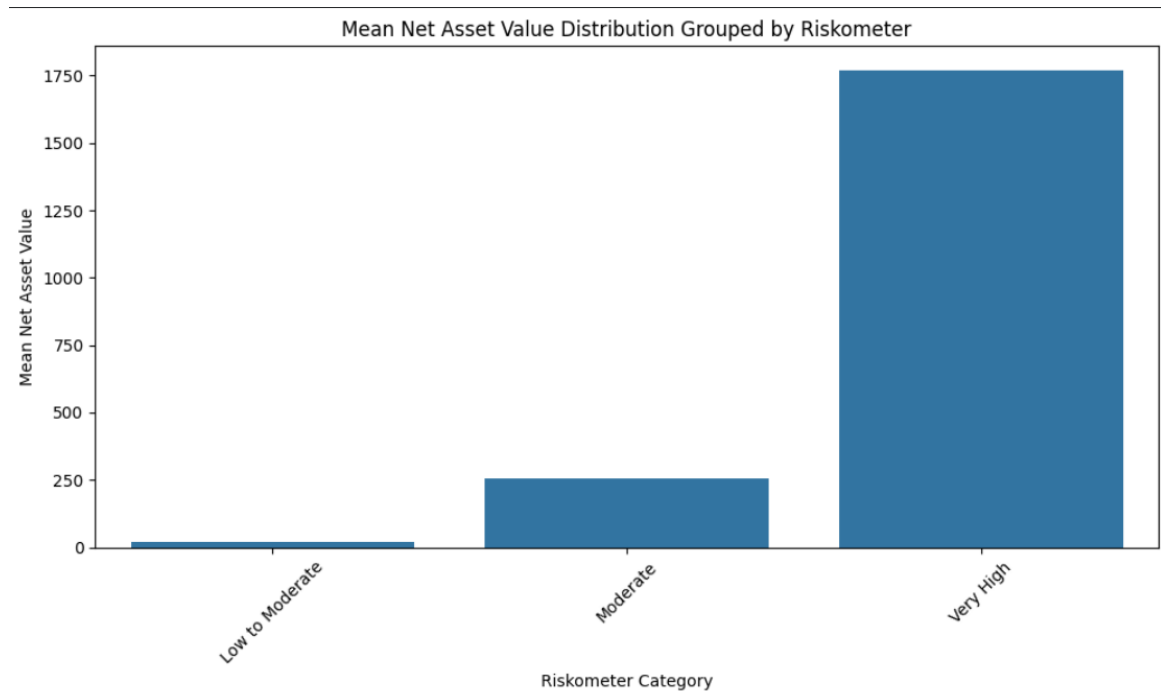


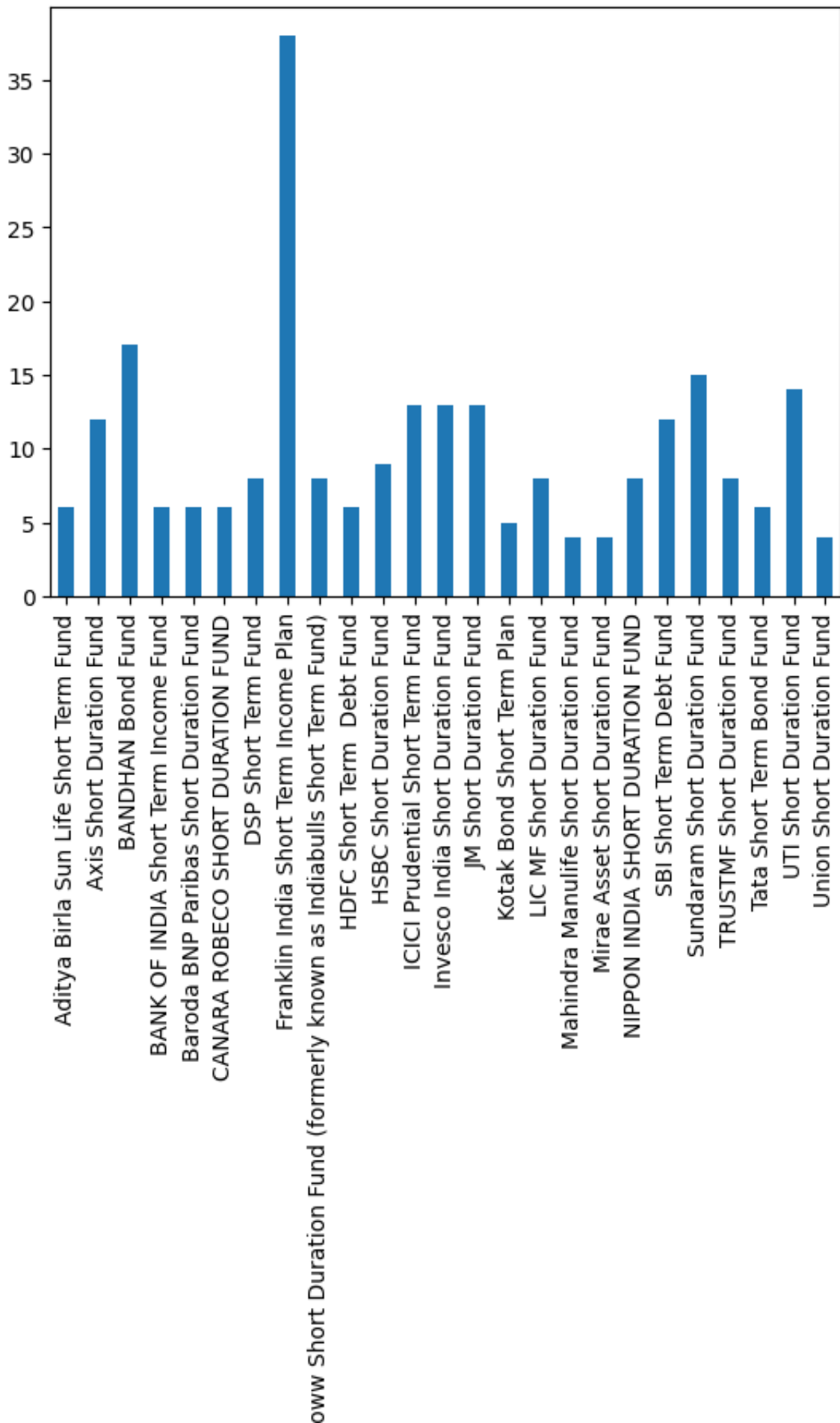
Figure 3.13: Mean NAV Grouped by Riskometer

This bar plot shows the mean Net Asset Value (NAV) for each Riskometer category. Funds classified under Very High risk have the highest mean NAV by a large margin. This clearly shows that High risk funds are often attractive.

Meanwhile, Low to Moderate risk schemes have the lowest mean NAV.



### 3.2.7 Distribution of NAV data points for main funds



This bar plot shows the number of NAV entries available for each fund. We obtained this by grouping the dataset by Fund Name and counting the corresponding Scheme NAV Name records. The height of each bar represents how many NAV data points exist for that particular scheme. Funds with taller bars have more historical NAV entries. This indicates better data availability and tracking. On the other hand, shorter bars indicate fewer recorded NAV values.

### 3.3 Individual banks

#### 3.3.1 Time-series trend of NAV Values for SBI

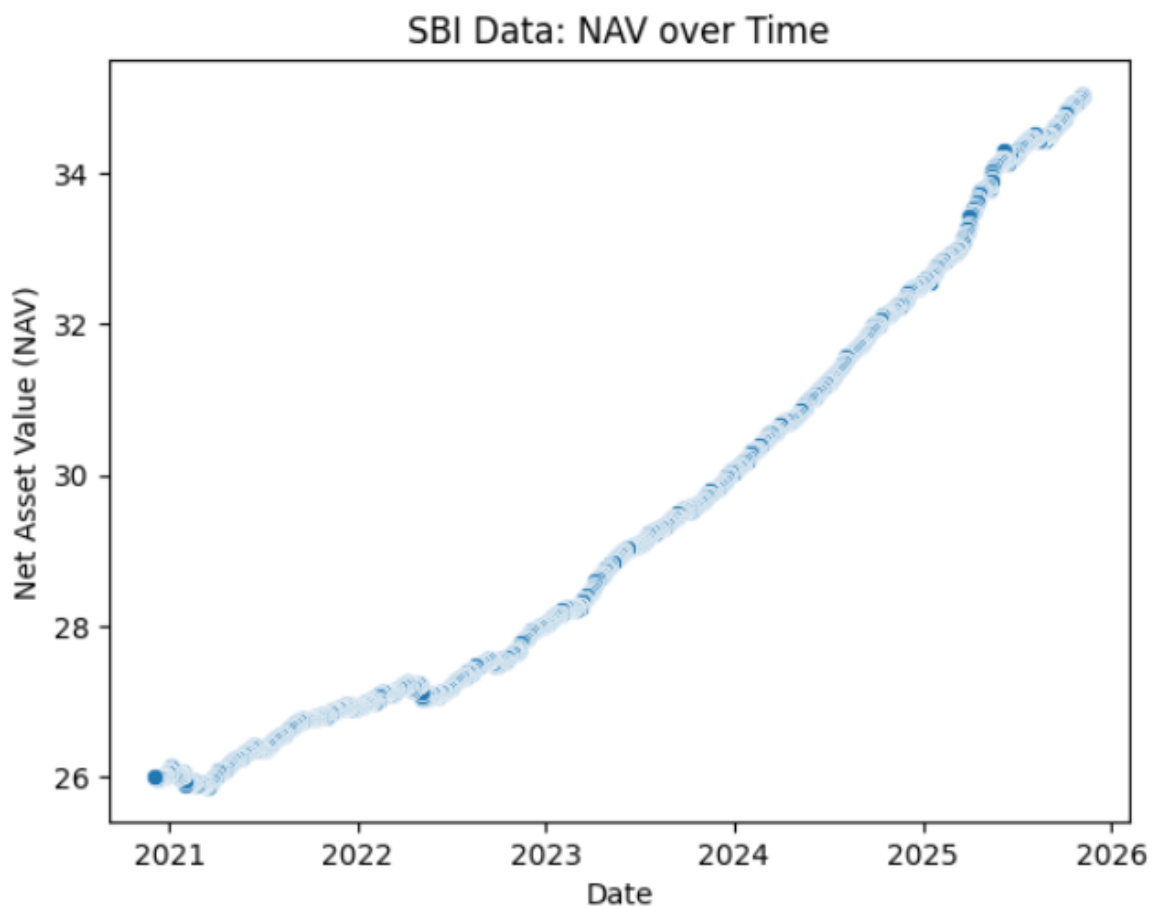


Figure 3.15: SBI Data: NAV over time

This is a scatter plot showing the time-series trend of NAV values for the SBI fund.

Even though it is a scatter plot, the points form a clear upward trend, which visually looks like a smooth line because NAV is changing gradually over time.

The plotted time series represents the Net Asset Value (NAV) of the SBI Short Duration Debt Fund observed from 2021 to 2025. The chart shows a consistently upward trend. This shows a steady growth in the fund's value over time. Starting around 26 Rs in early 2021, the NAV gradually increases with minor short-term fluctuations caused by daily market movements.

### 3.3.2 Time-series trend of NAV Values for Axis Bank

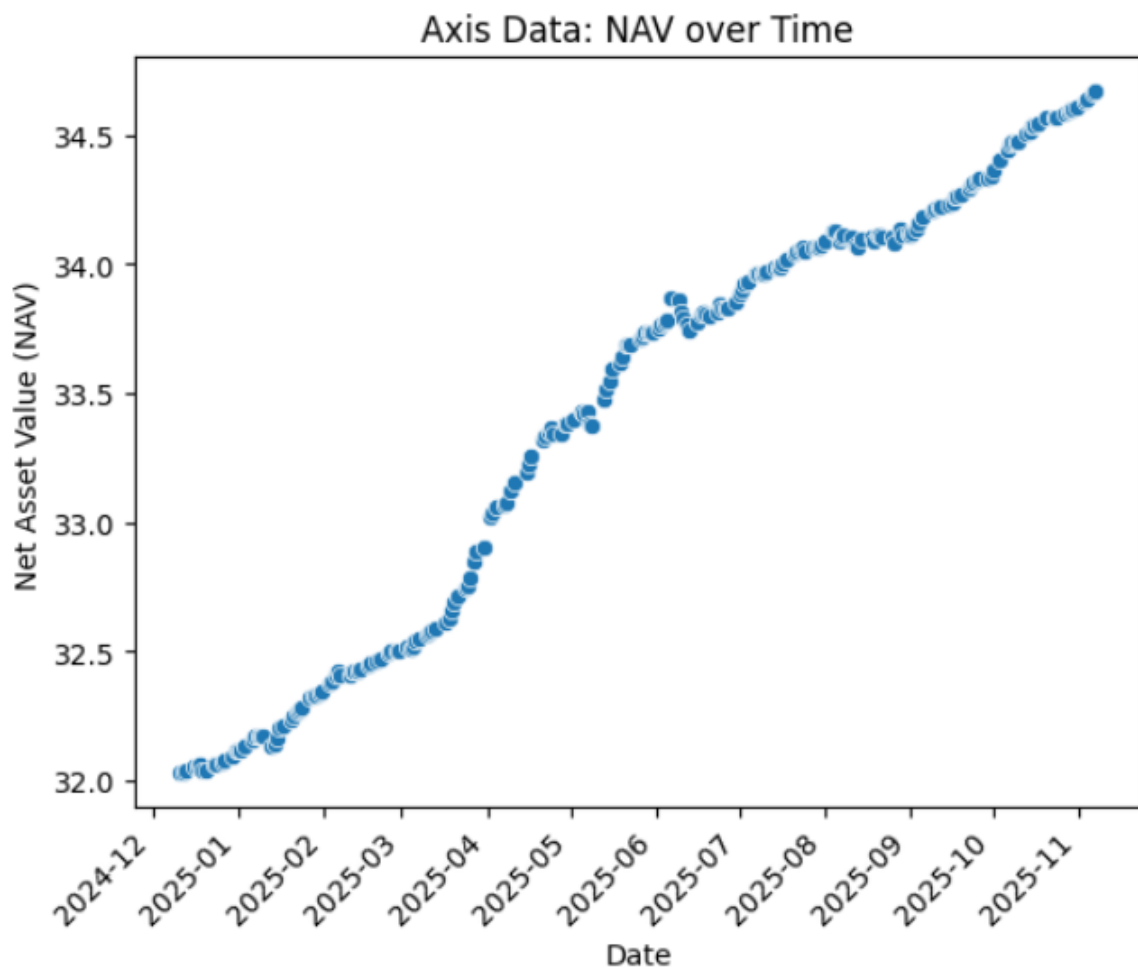


Figure 3.16: Axis Data: NAV over time

This is a scatter plot showing the time-series trend of NAV values for the Axis Bank fund. The scatter plot illustrates the Net Asset Value (NAV) movement of the Axis Short Duration Debt Fund from late 2024 to late 2025. The NAV starts at around 32 Rs. and shows a consistent upward trend throughout the year. This reflects steady growth in the fund's underlying debt instruments. The curve rises gradually from December 2024 through early 2025, indicating stable market conditions. A notable acceleration in NAV growth occurs around April to June 2025, suggesting a favorable period for bond valuations or interest rate conditions during these months.

### 3.3.3 Time-series trend of NAV Values for HDFC Bank

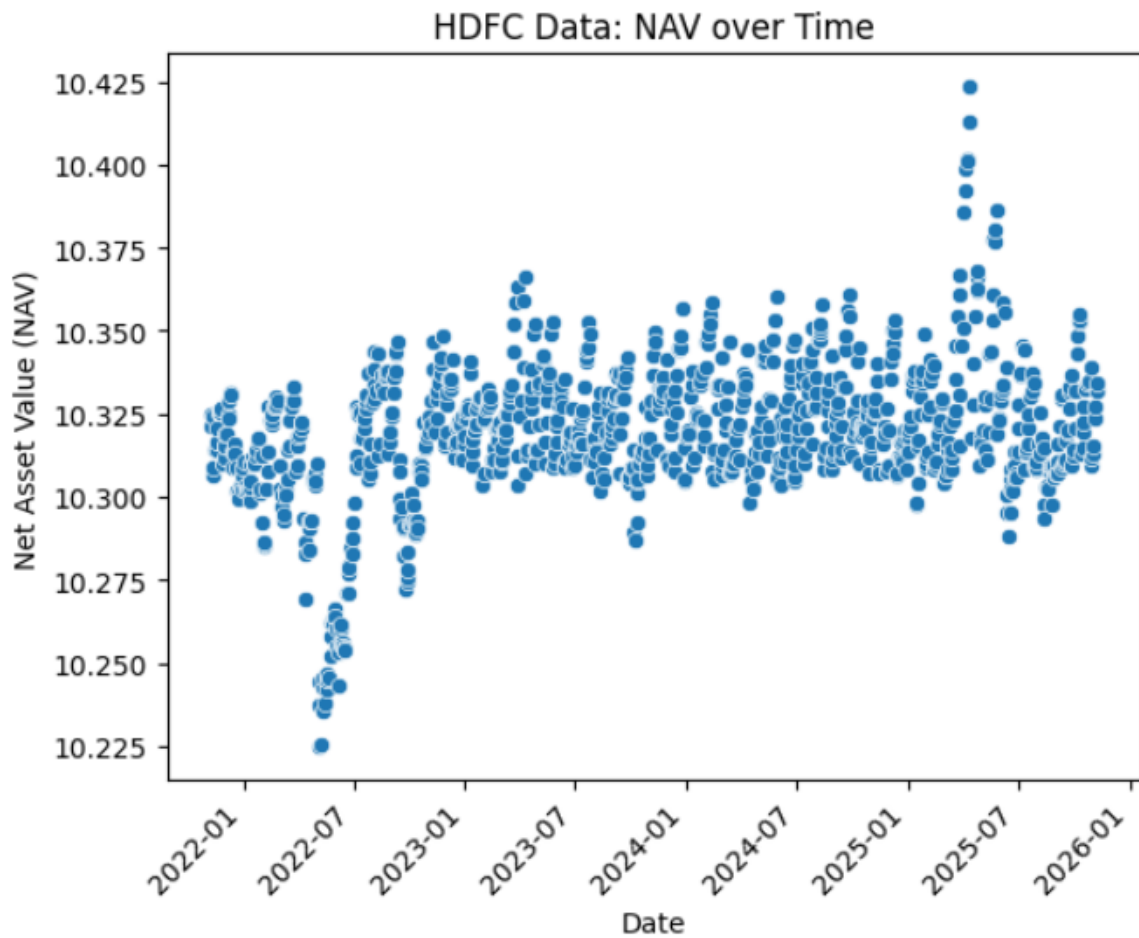


Figure 3.17: HDFC: NAV over time

This is a scatter plot showing the time-series trend of NAV values for the Axis Bank fund. Unlike the smooth upward trend seen in SBI and Axis funds, this fund displays more frequent fluctuations around a narrow NAV range. The NAV oscillates mostly between 10.25 Rs. and 10.35 Rs, indicating that the fund maintains a stable but tightly bounded value over time.

# Chapter 4. Feature Engineering

## 4.1 Feature selection

Feature selection is the process of identifying the most relevant features for the model, helping to improve both performance and interpretability. It also helps in reducing overfitting by eliminating irrelevant or redundant features.

### 4.1.1 Feature Selection for Classification Model

In this project, we have decided to use a Decision Tree Classifier to build the prediction model. We had a dataset named schemes which had six columns in it. The six columns are as follows:

- Scheme NAV Name
- Net Asset Value
- Rating
- Riskometer
- Date
- Launch Date

Out of this, there were two date columns which were not useful because launch dates do not influence the prediction task of Riskometer. Scheme names were also excluded, as they do not contribute meaningful numerical or categorical patterns. The remaining relevant features were as follows:

- Net Asset Value
- Rating

Our target variable is Riskometer, a categorical column representing fund risk levels.

### 4.1.2 Feature Selection for Regression Model

For the SBI and Axis funds, the NAV time-series plots showed an almost linear trend. This shows a near-linear relationship between time and NAV. Because of this pattern, we trained a Linear Regression model to capture this. The dataset for regression contained only two relevant columns:

- Date

- Net Asset Value (NAV)

Here, Date was used as the feature (independent variable). The NAV column is the target variable to be predicted. This simple structure allowed the regression model to learn how NAV changes over time in a continuous manner. This method seemed proper as NAV showed linear relation with Dates.



# Chapter 5. Model fitting

## 5.1 Decision Tree Classifier

**Goal:** We will predict risk which has three categories Low to Moderate, Moderate, Very high by Net Asset value and Rating.

### 5.1.1 Data Splitting:

After identifying the most important features, the dataset was split into training (80%) and testing (20%) sets to ensure the model could learn from most of the data while preserving a separate portion for testing its performance. Using a 80-20 split helps achieve a balance between training the model effectively and accurately assessing how well it generalizes to new data.

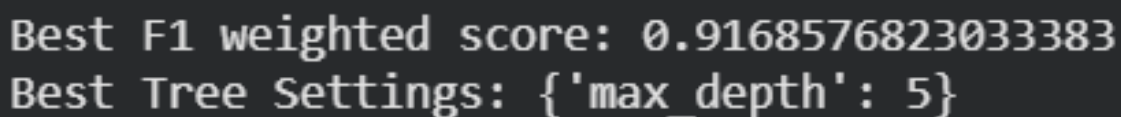
### 5.1.2 Model Training:

- **Why Decision Tree?**

We selected the Decision Tree Classifier because it is highly effective for datasets that contain a mix of numerical and categorical features. Our target variable, the Riskometer, is a categorical label, and decision trees naturally handle categorical output without requiring additional encoding. The model also provides clear interpretability. It allows us to understand the decision rules behind each prediction, which is good for analysis related to finance.

- **Hyper parameters:** The main hyperparameters used in Decision Tree model is:

- **max\_depth:** This determines how deep the decision tree is allowed to grow. We have used GridSearchCV, which allows us to find the best-forming depth of the tree, which came 5 in our case. We have fine-tuned this parameter.



```
Best F1 weighted score: 0.9168576823033383
Best Tree Settings: {'max_depth': 5}
```

Figure 5.1: Name vs AUM

- **random.state:** Although it is not part of tuning, it is still a hyperparameter which controls randomness in data splitting and tree building. The default value is 42. We use

42 simply because it makes results reproducible and is widely accepted convention in Machine Learning,

### 5.1.3 Model Testing:

- **Evaluation Metric:** For model tuning using GridSearchCV, we used the F1-weighted score as our evaluation metric. The F1 score is the harmonic mean of precision and recall, making it effective for handling imbalanced classes. The weighted version ensures that each class contributes proportionally based on its frequency. It prevents the model from simply predicting the majority class. Along with F1-weighted, we also calculated accuracy on the selected best model to validate its overall correctness.
- **Performance:**  
The best model achieved an F1-weighted score of 91% at a max\_depth of 5, indicating strong balance between precision and recall across all Riskometer categories. Additionally, the model delivered an accuracy of 94% on the test data.

## 5.2 Regression model

**Goal:** We will predict NAV value from date for both sbi and axis bank fund.

We converted date column into numerical so the regression model could treat dates as continuous numeric values for prediction.

### 5.2.1 Model Training:

- **Why linear regression:** We observed the graphs between dates and Net Asset Value in the datasets of SBI and Axis Bank, which showed strong linear relationship between time and NAV. Therefore, applying Linear Regression became a suitable and justified approach to model and predict future NAV values.

### 5.2.2 Model Testing:

- **Evaluation Metric:** The R2 score is an easy-to-understand and highly interpretable evaluation metric used for regression models. It measures how well the independent variable (Date) explains the variance in the target variable (Net Asset Value). A higher R2 value indicates that the model is able to capture more of the underlying trend.
- **Performance:** We used 5-fold cross-validation to ensure proper evaluation of the regression model. In this method, the dataset is split into five equal parts, and in each cycle, four parts are used for training while the remaining one part is used for testing. This process is repeated five times. This ensures that every fund is used exactly once as a test case. This helps to reduce bias and provide a more stable estimate of model performance. We found average r2 score 0.95 which demonstrates effectiveness of our model.

# Chapter 6. Conclusion & future scope

## 6.1 Findings/observations

Through an extensive exploratory data analysis of the Short Duration Debt Funds, we got following things out of it:

- These are short term debt funds, if we invest for short duration (1-3 years), as it is showed in [3.10](#), then we get good amount of returns which has been a general trend for all funds. If the duration increases to 5-7 years, then the returns decreases.
- The majority of returns are between 7-9% between 1-2 years, which has been a general trend for all funds as we can find in [3.1](#)
- From the dendogram given in [3.9](#), we can find that top most companies in terms of Asset Management like SBI, HDFC, Axis Bank are highly correlated and they employ almost similar strategies, so if an investor holds about HDFC and SBI short term debt funds, he/she is likely to achieve same benefits.
- From [3.5](#), we can see that most of the main funds that we have done Exploratory Data Analysis on have started or launched around 2013-14 only. This indicates that most of the active funds in this category entered the market around the same time. It may be due to increased demand for stable short-term investment products during that period.

## 6.2 Challenges

- In the missing values in the dataset that we downloaded, in the place of null values, there were dashes in the excel file, which led to a great problem while converting the data type, dash ('-') is considered as string and not as a null value.
- We found difficulty in gathering of data as there are only 26 funds. Then we only found 249 rows of short duration schemes by this funds. But the data we gathered had not sufficient number of columns for detailed analysis. Hence, we also found data for individual banks which had many rows but few columns only.
- As mentioned in above point, there was limited data so we found very difficulty in training our model and make it robust and find best parameter for it.
- Ensuring that the insights derived from our data visualizations and model were both meaningful and easily interpretable for investors and financial decision-makers posed a unique challenge in the context of short-duration debt funds.

## 6.3 Future plan

### 6.3.1 Model Refinement

We plan to refine the model by using more datasets with some additional features and with additional algorithms such as Random Forest or Gradient boosting for improved accuracy.

### 6.3.2 Real Time Application

A real-time system can be developed to continuously update fund metrics and automatically predict the Riskometer category as new NAV and rating data becomes available. This will help investors to receive timely risk insights.

### 6.3.3 Expanding Scope

The analysis can be expanded to include other debt funds such as liquid funds and overnight funds. This is done for broader industry coverage. Additional features like market indicators, interest rate trends, and credit spreads can also be integrated to get some depth which is useful.

# Group Contribution

## **Pranav Bavadiya (202301458)**

Data Collection, Data Cleaning, Visualizations, Model Training, Report Writing

## **Om Patel (202301027)**

Data Collection, Data Cleaning, Visualizations, Model Training, Report Writing

## **Chirag Katkoriya (202301259)**

Data Collection, Data Cleaning, Visualizations, Model Training, Report Writing,

# Short Bio

1. **Pranav Bavadiya** is a passionate software engineer with expertise in data analysis. He is pursuing a Bachelor's degree in Information and Communication Technology with minors in Computational Science from Dhirubhai Ambani University and has worked on various projects involving data analysis.

Throughout his career, he has actively contributed to many projects and has a keen interest in exploring new technologies. He is proficient in programming languages such as C++ and Python and enjoys tackling challenging problems in software development.

In addition to his technical skills, he is an effective team player with excellent communication skills. He enjoys collaborating with colleagues and has a track record of delivering solutions within tight deadlines.

Outside of work, he enjoys traveling, playing the guitar, listening to good music and volunteering at local coding workshops to inspire the next generation of developers. In future he wants to become a scientist who will find something which people dare to imagine.

2. **Om Patel** is a passionate software engineer with expertise in web development and data analysis. He is pursuing Bachelors in Information and Communication Technology from Dhirubhai Ambani University and has worked on various projects involving front-end and back-end development.

Throughout his career, Om has actively contributed to many projects and has a keen interest

in exploring new technologies. He is proficient in programming languages such as C++, Python and JavaScript. He enjoys tackling challenging problems in software development.

In addition to his technical skills, Om is an effective team player with excellent communication skills. He enjoys collaborating with colleagues and has a track record of delivering high-quality solutions within tight deadlines.

Outside of work, Om enjoys watching cricket, public speaking and volunteering at local coding workshops to inspire the next generation of developers.

3. **Chirag Katkoriya** is a passionate software engineer with expertise in web development and interest in exploratory data analysis. He is pursuing Bachelor of Technology in Information and Communication Technology from Dhirubhai Ambani University.

Throughout his career, Chirag has actively contributed to projects and has a keen interest in exploring new technologies. He is proficient in programming languages such as C++, JavaScript, and Python and enjoys tackling challenging problems in software development.

In addition to his technical skills, he is an effective team player with excellent communication skills. He enjoys collaborating with colleagues and has a track record of delivering quality solutions.

Outside of work, he enjoys watching cricket, coding, public speaking and volunteering at social workshops.

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- [5] SBI Data Url: <https://dhan.co/mutual-funds/sbi-short-term-debt-fund-direct-growth>